

Algebraic Geometry 4

Lecture notes with examples

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1 Construction of $\text{Proj } S$

Problem statement

We check the details of the construction of Proj in lecture. Recall, in analogy with $\text{Spec } A$, an affine scheme, we can define a projective scheme.

Let

$$S = \bigoplus_{d=0}^{\infty} S_d$$

be a graded ring. We will define $\text{Proj } S$.

(i) We write

$$S_+ = \bigoplus_{d=1}^{\infty} S_d,$$

the irrelevant ideal. Define $\text{Proj } S$ to be the set of all homogeneous prime ideals of S not containing the irrelevant ideal.

If $I \subseteq S$ is a homogeneous ideal, let $V(I)$ denote the set of all primes in $\text{Proj } S$ containing I . Show these form the closed sets of a topology on $\text{Proj } S$.

(ii) We define a sheaf $\mathcal{O}$ on $\text{Proj } S$. For $\mathfrak{p} \in \text{Proj } S$, let $S_{(\mathfrak{p})}$ be the set of elements of the localization $S_{\mathfrak{p}}$ which are homogeneous of degree zero, that is, a ratio of two elements of S of the same degree.

For $U \subseteq \text{Proj } S$ open, define $\mathcal{O}(U)$ to be the set of functions

$$s : U \longrightarrow \prod_{\mathfrak{p} \in U} S_{(\mathfrak{p})}$$

such that $s(\mathfrak{p}) \in S_{(\mathfrak{p})}$, and every point $\mathfrak{p} \in U$ has an open neighbourhood V for which there exist homogeneous $f, g \in S$ of the same degree, with

$$g \notin \mathfrak{q}$$

for all $\mathfrak{q} \in V$, such that

$$s(\mathfrak{q}) = \frac{f}{g}$$

for all $\mathfrak{q} \in V$.

Then show:

(a) The stalk of $\mathcal{O}$ at $\mathfrak{q}$ is

$$\mathcal{O}_{\mathfrak{q}} \cong S_{(\mathfrak{q})}.$$

(b) For any homogeneous $f \in S_+$, let $D_+(f)$ be the set of primes of $\text{Proj } S$ not containing f . Show the sets $D_+(f)$ cover $\text{Proj } S$, and for each such open set there is an isomorphism of locally ringed spaces

$$(D_+(f), \mathcal{O}|_{D_+(f)}) \cong \text{Spec } S_{(f)}.$$

Here $S_{(f)}$ denotes the subring of elements of degree 0 in the localization S_f .

Hint. It is easy to define a map

$$\psi : D_+(f) \longrightarrow \text{Spec } S_{(f)}.$$

However, constructing its inverse θ is not so easy. Given $\mathfrak{q} \in \operatorname{Spec} S_{(f)}$, set

$$\mathfrak{q}_n := \left\{ s \in S_n : \frac{s^{\deg f}}{f^n} \in \mathfrak{q} \right\}.$$

Show that $\mathfrak{q}_n$ is closed under addition and that

$$\theta(\mathfrak{q}) = \bigoplus_n \mathfrak{q}_n$$

is a homogeneous prime ideal in $D_+(f)$.

(c) Conclude that $\operatorname{Proj} S$ is a scheme.

(d) Show that if k is an algebraically closed field, then the set of closed points of

$$\operatorname{Proj} k[x_0, \dots, x_n]$$

is in one-to-one correspondence with points of

$$(k^{n+1} \setminus \{0\})/k^*$$

with the usual action of k^* by scalar multiplication.

Show that if

$$I \subseteq k[x_0, \dots, x_n]$$

is a homogeneous ideal, then the closed points of

$$\operatorname{Proj} k[x_0, \dots, x_n]/I$$

are in one-to-one correspondence with equivalence classes of points

$$(a_0, \dots, a_n) \in (k^{n+1} \setminus \{0\})/k^*$$

such that

$$F(a_0, \dots, a_n) = 0$$

for all homogeneous $F \in I$.

Solution

Let

$$S = \bigoplus_{d \geq 0} S_d$$

be a graded ring. The irrelevant ideal is

$$S_+ = \bigoplus_{d \geq 1} S_d.$$

We define

$$\operatorname{Proj} S = \{ \mathfrak{p} \subset S : \mathfrak{p} \text{ is homogeneous prime and } S_+ \not\subset \mathfrak{p} \}.$$

1.1 The topology on $\text{Proj } S$

For a homogeneous ideal $I \subseteq S$, define

$$V(I) = \{\mathfrak{p} \in \text{Proj } S : I \subseteq \mathfrak{p}\}.$$

We prove that these subsets form the closed sets of a topology.

First,

$$V(0) = \text{Proj } S,$$

because every prime ideal contains 0.

Also,

$$V(S_+) = \emptyset,$$

because by definition no point of $\text{Proj } S$ contains S_+ .

Now let $\{I_\alpha\}_{\alpha \in A}$ be a family of homogeneous ideals. Then

$$\bigcap_{\alpha \in A} V(I_\alpha) = V\left(\sum_{\alpha \in A} I_\alpha\right).$$

Indeed, a prime ideal contains every I_α if and only if it contains their sum.

Thus arbitrary intersections of closed sets are closed.

Now let $I, J \subseteq S$ be homogeneous ideals. Then

$$V(I) \cup V(J) = V(IJ).$$

Indeed, if $\mathfrak{p} \supseteq I$, then $\mathfrak{p} \supseteq IJ$, and similarly if $\mathfrak{p} \supseteq J$.

Conversely, suppose

$$IJ \subseteq \mathfrak{p}.$$

If $I \not\subseteq \mathfrak{p}$, choose $a \in I$ with $a \notin \mathfrak{p}$. Then for every $b \in J$,

$$ab \in IJ \subseteq \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime and $a \notin \mathfrak{p}$, we get

$$b \in \mathfrak{p}.$$

Hence $J \subseteq \mathfrak{p}$. Therefore

$$\mathfrak{p} \in V(I) \cup V(J).$$

So finite unions of closed sets are closed.

Hence the sets

$$V(I)$$

for homogeneous ideals $I \subseteq S$ define the closed sets of a topology on $\text{Proj } S$.

1.2 Basic open subsets

For a homogeneous element $f \in S$, define

$$D_+(f) = \text{Proj } S \setminus V((f)).$$

Equivalently,

$$D_+(f) = \{\mathfrak{p} \in \text{Proj } S : f \notin \mathfrak{p}\}.$$

These are the projective analogues of the affine basic open sets

$$D(f) \subseteq \text{Spec } A.$$

They satisfy

$$D_+(f) \cap D_+(g) = D_+(fg).$$

Indeed,

$$\mathfrak{p} \in D_+(f) \cap D_+(g)$$

if and only if

$$f \notin \mathfrak{p} \quad \text{and} \quad g \notin \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime, this is equivalent to

$$fg \notin \mathfrak{p}.$$

Hence

$$D_+(f) \cap D_+(g) = D_+(fg).$$

The sets $D_+(f)$, where $f \in S_+$ is homogeneous, cover $\text{Proj } S$. Indeed, if

$$\mathfrak{p} \in \text{Proj } S,$$

then

$$S_+ \not\subseteq \mathfrak{p}.$$

So there exists a homogeneous element

$$f \in S_+$$

such that

$$f \notin \mathfrak{p}.$$

Thus

$$\mathfrak{p} \in D_+(f).$$

Therefore the $D_+(f)$ cover $\text{Proj } S$.

1.3 The rings $S_{(\mathfrak{p})}$

Let $\mathfrak{p} \in \text{Proj } S$. We define

$$S_{(\mathfrak{p})}$$

to be the degree-zero part of the homogeneous localization of S at $\mathfrak{p}$.

Concretely,

$$S_{(\mathfrak{p})} = \left\{ \frac{a}{b} : a, b \in S \text{ homogeneous, } \deg a = \deg b, b \notin \mathfrak{p} \right\}.$$

This is a local ring. Its maximal ideal is

$$\mathfrak{p}_{(\mathfrak{p})} = \left\{ \frac{a}{b} \in S_{(\mathfrak{p})} : a \in \mathfrak{p} \right\}.$$

Indeed, if

$$\frac{a}{b} \notin \mathfrak{p}_{(\mathfrak{p})},$$

then

$$a \notin \mathfrak{p}.$$

Since also $b \notin \mathfrak{p}$, the inverse

$$\frac{b}{a}$$

is defined in $S_{(\mathfrak{p})}$. Therefore every element outside $\mathfrak{p}_{(\mathfrak{p})}$ is invertible. Hence $S_{(\mathfrak{p})}$ is local.

1.4 The structure sheaf

For an open subset $U \subseteq \text{Proj } S$, the ring $\mathcal{O}(U)$ consists of functions

$$s : U \longrightarrow \prod_{\mathfrak{p} \in U} S_{(\mathfrak{p})}$$

such that

$$s(\mathfrak{p}) \in S_{(\mathfrak{p})}$$

for each $\mathfrak{p} \in U$, and such that s is locally represented by a fraction

$$\frac{f}{g}$$

where $f, g \in S$ are homogeneous of the same degree.

This is exactly the projective analogue of the structure sheaf on $\text{Spec } A$, except that only degree-zero homogeneous fractions are allowed.

1.5 The stalk of $\mathcal{O}$ at a point

We prove that

$$\mathcal{O}_{\mathfrak{q}} \cong S_{(\mathfrak{q})}.$$

Define

$$\Phi : \mathcal{O}_{\mathfrak{q}} \longrightarrow S_{(\mathfrak{q})}$$

by evaluation at $\mathfrak{q}$. If a germ is represented by a section $s \in \mathcal{O}(U)$, with $\mathfrak{q} \in U$, define

$$\Phi(s_{\mathfrak{q}}) = s(\mathfrak{q}).$$

This is a ring homomorphism.

We show that Φ is surjective. Let

$$\frac{a}{b} \in S_{(\mathfrak{q})}.$$

Then a, b are homogeneous of the same degree and

$$b \notin \mathfrak{q}.$$

Hence

$$\mathfrak{q} \in D_+(b).$$

Define a section on $D_+(b)$ by

$$s(\mathfrak{p}) = \frac{a}{b}.$$

This is regular by definition. Its germ at $\mathfrak{q}$ maps to a/b . Therefore Φ is surjective.

Now suppose

$$\Phi(s_{\mathfrak{q}}) = 0.$$

Since s is locally a homogeneous fraction, after shrinking around $\mathfrak{q}$ we may write

$$s(\mathfrak{p}) = \frac{a}{b}$$

on an open neighbourhood V of $\mathfrak{q}$, where a, b are homogeneous of the same degree and $b \notin \mathfrak{p}$ for all $\mathfrak{p} \in V$.

The equality

$$s(\mathfrak{q}) = 0$$

means

$$\frac{a}{b} = 0$$

in $S_{(\mathfrak{q})}$. Hence there exists a homogeneous element

$$h \notin \mathfrak{q}$$

such that

$$ha = 0.$$

Now restrict to

$$V \cap D_+(h).$$

On this smaller neighbourhood, $h \notin \mathfrak{p}$, so

$$\frac{a}{b} = 0$$

in $S_{(\mathfrak{p})}$ for every $\mathfrak{p} \in V \cap D_+(h)$. Hence the germ $s_{\mathfrak{q}}$ is zero.

Therefore Φ is injective and surjective. Thus

$$\boxed{\mathcal{O}_{\mathfrak{q}} \cong S_{(\mathfrak{q})}.$$

1.6 The affine open pieces

Let $f \in S_+$ be homogeneous and set

$$d = \deg f.$$

The localization S_f is naturally graded. Let

$$S_{(f)} = (S_f)_0$$

be its degree-zero part.

Explicitly,

$$S_{(f)} = \left\{ \frac{a}{f^m} : a \in S_{md} \right\}.$$

We prove that

$$(D_+(f), \mathcal{O}|_{D_+(f)}) \cong \text{Spec } S_{(f)}.$$

1.7 The map $D_+(f) \rightarrow \text{Spec } S_{(f)}$

Let

$$\mathfrak{p} \in D_+(f).$$

Then $f \notin \mathfrak{p}$, so we can localize $\mathfrak{p}$ in S_f . Define

$$\psi(\mathfrak{p}) = \mathfrak{p}S_f \cap S_{(f)}.$$

This is a prime ideal of $S_{(f)}$, because $\mathfrak{p}S_f$ is prime in S_f , and the contraction of a prime ideal is prime.

Therefore we have a map

$$\psi : D_+(f) \longrightarrow \text{Spec } S_{(f)}.$$

1.8 The inverse map

Now let

$$\mathfrak{q} \in \text{Spec } S_{(f)}.$$

We construct a homogeneous prime ideal of S .

For $n \geq 0$, define

$$\mathfrak{q}_n = \left\{ s \in S_n : \frac{s^d}{f^n} \in \mathfrak{q} \right\}.$$

This makes sense because if $s \in S_n$, then

$$\deg(s^d) = dn$$

and

$$\deg(f^n) = dn,$$

so

$$\frac{s^d}{f^n}$$

has degree zero and belongs to $S_{(f)}$.

Now set

$$\theta(\mathfrak{q}) = \bigoplus_{n \geq 0} \mathfrak{q}_n.$$

We show that $\theta(\mathfrak{q})$ is a homogeneous prime ideal.

First, by construction,

$$\mathfrak{q}_n \subseteq S_n.$$

Therefore $\theta(\mathfrak{q})$ is homogeneous.

Now we prove that each $\mathfrak{q}_n$ is closed under addition. Let

$$s, t \in \mathfrak{q}_n.$$

Then

$$\frac{s^d}{f^n} \in \mathfrak{q}, \quad \frac{t^d}{f^n} \in \mathfrak{q}.$$

We need to show that

$$s + t \in \mathfrak{q}_n,$$

that is,

$$\frac{(s+t)^d}{f^n} \in \mathfrak{q}.$$

By the binomial theorem,

$$\frac{(s+t)^d}{f^n} = \sum_{i=0}^d \binom{d}{i} \frac{s^i t^{d-i}}{f^n}.$$

Each summand has degree zero.

For each i , consider

$$\frac{s^i t^{d-i}}{f^n}.$$

Its d -th power is

$$\left(\frac{s^i t^{d-i}}{f^n} \right)^d = \left(\frac{s^d}{f^n} \right)^i \left(\frac{t^d}{f^n} \right)^{d-i}.$$

Since

$$\frac{s^d}{f^n}, \frac{t^d}{f^n} \in \mathfrak{q},$$

the right-hand side lies in $\mathfrak{q}$. Since $\mathfrak{q}$ is prime, it is radical. Therefore

$$\frac{s^i t^{d-i}}{f^n} \in \mathfrak{q}.$$

Hence every term in the binomial expansion lies in $\mathfrak{q}$, so

$$\frac{(s+t)^d}{f^n} \in \mathfrak{q}.$$

Thus

$$s+t \in \mathfrak{q}_n.$$

Now we show that $\theta(\mathfrak{q})$ is an ideal. Let

$$s \in \mathfrak{q}_n$$

and let

$$r \in S_m$$

be homogeneous. Then

$$rs \in S_{m+n}.$$

Moreover,

$$\frac{(rs)^d}{f^{m+n}} = \frac{r^d}{f^m} \cdot \frac{s^d}{f^n}.$$

The first factor belongs to $S_{(f)}$, and the second factor belongs to $\mathfrak{q}$. Therefore

$$\frac{(rs)^d}{f^{m+n}} \in \mathfrak{q}.$$

Hence

$$rs \in \mathfrak{q}_{m+n}.$$

So $\theta(\mathfrak{q})$ is a homogeneous ideal.

Now we prove that $\theta(\mathfrak{q})$ is prime. Let $a \in S_m$ and $b \in S_n$ be homogeneous, and suppose

$$ab \in \theta(\mathfrak{q}).$$

Then

$$ab \in \mathfrak{q}_{m+n}.$$

Hence

$$\frac{(ab)^d}{f^{m+n}} \in \mathfrak{q}.$$

But

$$\frac{(ab)^d}{f^{m+n}} = \frac{a^d}{f^m} \cdot \frac{b^d}{f^n}.$$

Since $\mathfrak{q}$ is prime, either

$$\frac{a^d}{f^m} \in \mathfrak{q}$$

or

$$\frac{b^d}{f^n} \in \mathfrak{q}.$$

Thus either

$$a \in \mathfrak{q}_m$$

or

$$b \in \mathfrak{q}_n.$$

Therefore either

$$a \in \theta(\mathfrak{q})$$

or

$$b \in \theta(\mathfrak{q}).$$

So $\theta(\mathfrak{q})$ is prime.

Finally,

$$f \notin \theta(\mathfrak{q}).$$

Indeed, $f \in S_d$, and

$$\frac{f^d}{f^d} = 1 \notin \mathfrak{q}.$$

So

$$f \notin \theta(\mathfrak{q}).$$

Therefore

$$\theta(\mathfrak{q}) \in D_+(f).$$

Thus we have constructed a map

$$\theta : \operatorname{Spec} S_{(f)} \longrightarrow D_+(f).$$

1.9 The two maps are inverse

We now show that

$$\psi \circ \theta = \operatorname{id} \quad \text{and} \quad \theta \circ \psi = \operatorname{id}.$$

Let

$$\mathfrak{q} \in \operatorname{Spec} S_{(f)}.$$

An element of $S_{(f)}$ has the form

$$\frac{a}{f^m},$$

where $a \in S_{md}$. Then

$$\frac{a}{f^m} \in \psi(\theta(\mathfrak{q}))$$

if and only if

$$a \in \theta(\mathfrak{q}).$$

By definition of θ , this is equivalent to

$$\frac{a^d}{f^{md}} \in \mathfrak{q}.$$

But

$$\frac{a^d}{f^{md}} = \left(\frac{a}{f^m} \right)^d.$$

Since $\mathfrak{q}$ is prime, hence radical, this is equivalent to

$$\frac{a}{f^m} \in \mathfrak{q}.$$

Therefore

$$\psi(\theta(\mathfrak{q})) = \mathfrak{q}.$$

Conversely, let

$$\mathfrak{p} \in D_+(f).$$

For a homogeneous element $s \in S_n$, we have

$$s \in \theta(\psi(\mathfrak{p}))$$

if and only if

$$\frac{s^d}{f^n} \in \psi(\mathfrak{p}).$$

This is equivalent to

$$\frac{s^d}{f^n} \in \mathfrak{p}S_f.$$

Since $f \notin \mathfrak{p}$, this is equivalent to

$$s^d \in \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime, this is equivalent to

$$s \in \mathfrak{p}.$$

Hence

$$\theta(\psi(\mathfrak{p})) = \mathfrak{p}.$$

Therefore

$$D_+(f) \cong \operatorname{Spec} S_{(f)}$$

as topological spaces.

1.10 Compatibility with the structure sheaves

The homeomorphism above also identifies the structure sheaves.

Indeed, on $D_+(f)$, sections of $\mathcal{O}_{\operatorname{Proj} S}$ are locally represented by fractions

$$\frac{a}{b}$$

where a, b are homogeneous of the same degree.

On $\text{Spec } S_{(f)}$, sections of the affine structure sheaf are locally represented by fractions in localizations of $S_{(f)}$.

These two descriptions agree under the correspondence

$$D_+(f) \cong \text{Spec } S_{(f)}.$$

More explicitly, if $g \in S$ is homogeneous, then

$$D_+(f) \cap D_+(g)$$

corresponds to the affine principal open subset

$$D\left(\frac{g^d}{f^{\deg g}}\right) \subseteq \text{Spec } S_{(f)}.$$

On these basic opens, both sheaves are described by the same degree-zero localized fractions.

Therefore we obtain an isomorphism of locally ringed spaces

$$\boxed{(D_+(f), \mathcal{O}|_{D_+(f)}) \cong \text{Spec } S_{(f)}}.$$

1.11 Proj S is a scheme

The open subsets

$$D_+(f), \quad f \in S_+ \text{ homogeneous,}$$

cover $\text{Proj } S$.

For each such f , we have shown that

$$D_+(f) \cong \text{Spec } S_{(f)}$$

as locally ringed spaces.

Therefore $\text{Proj } S$ is covered by affine schemes. Hence

$$\boxed{\text{Proj } S \text{ is a scheme.}}$$

2 Closed points of projective space

Now let

$$S = k[x_0, \dots, x_n]$$

with the standard grading, and assume that k is algebraically closed.

Then

$$\text{Proj } k[x_0, \dots, x_n]$$

is projective n -space over k , denoted

$$\mathbb{P}_k^n.$$

We show that the closed points of $\mathbb{P}_k^n$ are in one-to-one correspondence with

$$(k^{n+1} \setminus \{0\})/k^*.$$

2.1 From a projective point to a homogeneous prime

Let

$$a = (a_0, \dots, a_n) \in k^{n+1} \setminus \{0\}.$$

The corresponding projective point is written

$$[a_0 : \dots : a_n].$$

Define

$$\mathfrak{p}_a = \{F \in k[x_0, \dots, x_n] \text{ homogeneous} : F(a_0, \dots, a_n) = 0\}.$$

Equivalently, $\mathfrak{p}_a$ is the kernel of the graded ring homomorphism

$$k[x_0, \dots, x_n] \longrightarrow k[t]$$

given by

$$x_i \longmapsto a_i t.$$

The kernel is homogeneous. It is prime because $k[t]$ is an integral domain.

Also, since $a \neq 0$, some $a_i \neq 0$. Hence

$$x_i \notin \mathfrak{p}_a.$$

Therefore

$$S_+ \not\subseteq \mathfrak{p}_a.$$

So

$$\mathfrak{p}_a \in \text{Proj } k[x_0, \dots, x_n].$$

If $b = \lambda a$ for some $\lambda \in k^*$, then for every homogeneous polynomial F of degree m ,

$$F(b) = F(\lambda a) = \lambda^m F(a).$$

Thus

$$F(a) = 0 \iff F(b) = 0.$$

Therefore

$$\mathfrak{p}_a = \mathfrak{p}_b.$$

So the homogeneous prime depends only on the equivalence class

$$[a_0 : \dots : a_n].$$

2.2 From a closed point to a projective point

Conversely, let

$$\mathfrak{p} \in \text{Proj } k[x_0, \dots, x_n]$$

be a closed point.

Since

$$S_+ \not\subseteq \mathfrak{p},$$

there exists some i such that

$$x_i \notin \mathfrak{p}.$$

Thus

$$\mathfrak{p} \in D_+(x_i).$$

But

$$D_+(x_i) \cong \operatorname{Spec} k \left[\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i} \right].$$

This affine scheme is isomorphic to

$$\mathbb{A}_k^n.$$

Since k is algebraically closed, the closed points of $\mathbb{A}_k^n$ are exactly the k -rational points. Therefore $\mathfrak{p}$ corresponds to a maximal ideal of the form

$$\left(\frac{x_0}{x_i} - b_0, \dots, \frac{\widehat{x_i}}{x_i} - 1, \dots, \frac{x_n}{x_i} - b_n \right).$$

This gives the projective point

$$[b_0 : \dots : b_{i-1} : 1 : b_{i+1} : \dots : b_n].$$

Hence every closed point of $\operatorname{Proj} k[x_0, \dots, x_n]$ arises from a projective point.

Therefore

$$\boxed{|\operatorname{Proj} k[x_0, \dots, x_n]|_{\text{closed}} \cong (k^{n+1} \setminus \{0\})/k^*}.$$

3 Closed points of projective varieties

Let

$$I \subseteq k[x_0, \dots, x_n]$$

be a homogeneous ideal. Consider

$$\operatorname{Proj} k[x_0, \dots, x_n]/I.$$

The closed points of this scheme are the closed points of projective space satisfying the equations in I .

Indeed, a point

$$[a_0 : \dots : a_n] \in \mathbb{P}_k^n$$

corresponds to the homogeneous prime

$$\mathfrak{p}_a = \{F \in k[x_0, \dots, x_n] \text{ homogeneous} : F(a_0, \dots, a_n) = 0\}.$$

It lies in

$$\operatorname{Proj} k[x_0, \dots, x_n]/I$$

if and only if

$$I \subseteq \mathfrak{p}_a.$$

This is equivalent to saying that

$$F(a_0, \dots, a_n) = 0$$

for every homogeneous polynomial

$$F \in I.$$

Therefore

$$|\mathrm{Proj} k[x_0, \dots, x_n]/I|_{\mathrm{closed}} \cong \{[a_0 : \dots : a_n] \in \mathbb{P}_k^n : F(a_0, \dots, a_n) = 0 \text{ for all homogeneous } F \in I\}.$$

Equivalently, the closed points of

$$\mathrm{Proj} k[x_0, \dots, x_n]/I$$

are exactly the projective equivalence classes of nonzero points

$$(a_0, \dots, a_n) \in k^{n+1} \setminus \{0\}$$

satisfying all homogeneous equations in I .

4 Summary

The construction of $\mathrm{Proj} S$ is the projective analogue of the construction of $\mathrm{Spec} A$.

Affine case	Projective case
$\mathrm{Spec} A$	$\mathrm{Proj} S$
prime ideals	homogeneous prime ideals not containing S_+
$V(I)$	$V(I) \subseteq \mathrm{Proj} S$
$D(f)$	$D_+(f)$
A_f	$S_{(f)}$
$D(f) \cong \mathrm{Spec} A_f$	$D_+(f) \cong \mathrm{Spec} S_{(f)}$

The key local fact is

$$D_+(f) \cong \mathrm{Spec} S_{(f)}.$$

Since the sets $D_+(f)$ cover $\mathrm{Proj} S$, this proves that

$$\mathrm{Proj} S \text{ is a scheme.}$$

5 Theory of the Construction of $\mathrm{Proj} S$

The construction of $\mathrm{Proj} S$ is the projective analogue of the construction of $\mathrm{Spec} A$.

In affine algebraic geometry, one begins with an ordinary commutative ring A and forms

$$\mathrm{Spec} A,$$

the set of all prime ideals of A .

In projective algebraic geometry, one begins with a graded ring

$$S = \bigoplus_{d \geq 0} S_d$$

and forms

$$\text{Proj } S,$$

the set of homogeneous prime ideals of S which do not contain the irrelevant ideal.

Thus the basic analogy is:

Affine geometry	Projective geometry
A a ring	$S = \bigoplus_{d \geq 0} S_d$ a graded ring
$\text{Spec } A$	$\text{Proj } S$
prime ideals	homogeneous prime ideals
ordinary localization	degree-zero homogeneous localization

The key new features in the projective case are:

1. the grading of the ring,
2. the irrelevant ideal S_+ ,
3. the restriction to homogeneous prime ideals,
4. the use of degree-zero fractions in the structure sheaf.

5.1 Graded Rings

Definition 5.1.1. A ring S is called a *graded ring* if it is written as a direct sum of abelian groups

$$S = \bigoplus_{d \geq 0} S_d$$

such that multiplication respects degree:

$$S_m S_n \subseteq S_{m+n}.$$

An element $s \in S_d$ is called *homogeneous of degree d* .

Every element $s \in S$ can be written uniquely as a finite sum

$$s = s_0 + s_1 + \cdots + s_N,$$

where

$$s_i \in S_i.$$

The elements s_i are called the homogeneous components of s .

Example 5.1.2. Let

$$S = k[x_0, \dots, x_n].$$

The standard grading is given by

$$S_d = \{\text{homogeneous polynomials of degree } d\}.$$

Thus

$$S_0 = k,$$

S_1 = the vector space of linear forms,

S_2 = the vector space of quadratic forms,

and so on.

5.2 The Irrelevant Ideal

Definition 5.2.1. Let

$$S = \bigoplus_{d \geq 0} S_d$$

be a graded ring. The *irrelevant ideal* of S is

$$S_+ = \bigoplus_{d \geq 1} S_d.$$

So S_+ consists of all elements of positive degree.

In the standard graded polynomial ring

$$S = k[x_0, \dots, x_n],$$

we have

$$S_+ = (x_0, \dots, x_n).$$

This ideal is called the irrelevant ideal because it corresponds to the forbidden origin

$$(0, \dots, 0)$$

in affine space.

Projective space is built from nonzero vectors modulo scalar multiplication:

$$(a_0, \dots, a_n) \sim (\lambda a_0, \dots, \lambda a_n), \quad \lambda \in k^*.$$

The zero vector is not allowed. Therefore the ideal corresponding to the origin must be excluded.

5.3 Definition of $\text{Proj } S$

Definition 5.3.1. Let

$$S = \bigoplus_{d \geq 0} S_d$$

be a graded ring. Then

$$\text{Proj } S$$

is the set of all homogeneous prime ideals $\mathfrak{p} \subset S$ such that

$$S_+ \not\subset \mathfrak{p}.$$

Thus

$$\text{Proj } S = \{\mathfrak{p} \subset S : \mathfrak{p} \text{ is homogeneous prime and } S_+ \not\subset \mathfrak{p}\}.$$

The condition

$$S_+ \not\subseteq \mathfrak{p}$$

means that not every positive-degree homogeneous element lies in $\mathfrak{p}$.

Equivalently, there exists some homogeneous element

$$f \in S_+$$

such that

$$f \notin \mathfrak{p}.$$

This is the algebraic version of the projective condition that at least one homogeneous coordinate is nonzero.

5.4 Why Only Homogeneous Prime Ideals?

Projective geometry is invariant under scalar multiplication.

A point of projective n -space over k is an equivalence class

$$[a_0 : \cdots : a_n],$$

where

$$(a_0, \dots, a_n) \in k^{n+1} \setminus \{0\},$$

and

$$(a_0, \dots, a_n) \sim (\lambda a_0, \dots, \lambda a_n)$$

for every

$$\lambda \in k^*.$$

Homogeneous polynomials behave well under this equivalence relation.

If $F \in k[x_0, \dots, x_n]$ is homogeneous of degree d , then

$$F(\lambda a_0, \dots, \lambda a_n) = \lambda^d F(a_0, \dots, a_n).$$

Hence

$$F(a_0, \dots, a_n) = 0$$

if and only if

$$F(\lambda a_0, \dots, \lambda a_n) = 0.$$

Therefore homogeneous equations define well-defined subsets of projective space.

Non-homogeneous equations generally do not define well-defined projective subsets.

For example,

$$x_0 + 1 = 0$$

is not homogeneous, and the condition is not invariant under scaling.

On the other hand,

$$x_0^2 + x_1 x_2 = 0$$

is homogeneous, and hence defines a projective hypersurface.

Therefore the geometry of projective schemes must be built from homogeneous ideals and homogeneous prime ideals.

5.5 Closed Sets in $\text{Proj } S$

Let $I \subseteq S$ be a homogeneous ideal.

Define

$$V_+(I) = \{\mathfrak{p} \in \text{Proj } S : I \subseteq \mathfrak{p}\}.$$

These are the projective closed sets.

Proposition 5.1. *The sets*

$$V_+(I),$$

where $I \subseteq S$ is a homogeneous ideal, define the closed sets of a topology on $\text{Proj } S$.

Proof. First,

$$V_+(0) = \text{Proj } S,$$

because every prime ideal contains 0.

Also,

$$V_+(S_+) = \emptyset,$$

because by definition no point of $\text{Proj } S$ contains S_+ .

Let $\{I_\alpha\}_{\alpha \in A}$ be a family of homogeneous ideals. Then

$$\bigcap_{\alpha \in A} V_+(I_\alpha) = V_+\left(\sum_{\alpha \in A} I_\alpha\right).$$

Indeed, a prime ideal contains every I_α if and only if it contains their sum.

Thus arbitrary intersections of closed sets are closed.

Now let $I, J \subseteq S$ be homogeneous ideals. Then

$$V_+(I) \cup V_+(J) = V_+(IJ).$$

If $\mathfrak{p} \in V_+(I) \cup V_+(J)$, then either

$$I \subseteq \mathfrak{p}$$

or

$$J \subseteq \mathfrak{p}.$$

In either case,

$$IJ \subseteq \mathfrak{p}.$$

Thus

$$\mathfrak{p} \in V_+(IJ).$$

Conversely, suppose

$$IJ \subseteq \mathfrak{p}.$$

If

$$I \not\subseteq \mathfrak{p},$$

choose

$$a \in I$$

with

$$a \notin \mathfrak{p}.$$

For every $b \in J$, we have

$$ab \in IJ \subseteq \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime and $a \notin \mathfrak{p}$, it follows that

$$b \in \mathfrak{p}.$$

Hence

$$J \subseteq \mathfrak{p}.$$

Therefore

$$\mathfrak{p} \in V_+(I) \cup V_+(J).$$

Thus finite unions of closed sets are closed.

Therefore the sets $V_+(I)$ define the closed sets of a topology on $\text{Proj } S$. $\square$

5.6 Basic Open Sets

For a homogeneous element $f \in S$, define

$$D_+(f) = \text{Proj } S \setminus V_+(f).$$

Equivalently,

$$D_+(f) = \{\mathfrak{p} \in \text{Proj } S : f \notin \mathfrak{p}\}.$$

These sets are the projective analogues of the affine basic open sets

$$D(f) \subseteq \text{Spec } A.$$

Proposition 5.2. *If $f, g \in S$ are homogeneous, then*

$$D_+(f) \cap D_+(g) = D_+(fg).$$

Proof. Let $\mathfrak{p} \in \text{Proj } S$. Then

$$\mathfrak{p} \in D_+(f) \cap D_+(g)$$

if and only if

$$f \notin \mathfrak{p} \quad \text{and} \quad g \notin \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime, this is equivalent to

$$fg \notin \mathfrak{p}.$$

Therefore

$$\mathfrak{p} \in D_+(fg).$$

Hence

$$D_+(f) \cap D_+(g) = D_+(fg).$$

$\square$

Proposition 5.3. *The sets $D_+(f)$, where $f \in S_+$ is homogeneous, cover $\text{Proj } S$.*

Proof. Let

$$\mathfrak{p} \in \text{Proj } S.$$

By definition,

$$S_+ \not\subseteq \mathfrak{p}.$$

Therefore there exists some homogeneous element

$$f \in S_+$$

such that

$$f \notin \mathfrak{p}.$$

Hence

$$\mathfrak{p} \in D_+(f).$$

Therefore the sets $D_+(f)$ cover $\text{Proj } S$. $\square$

5.7 Degree-Zero Localizations

The essential difference between $\text{Spec } A$ and $\text{Proj } S$ appears in the local rings.

In affine geometry, if $\mathfrak{p} \in \text{Spec } A$, then the local ring is

$$A_{\mathfrak{p}}.$$

In projective geometry, if $\mathfrak{p} \in \text{Proj } S$, we do not take all of $S_{\mathfrak{p}}$. Instead, we take only the degree-zero part.

Definition 5.7.1. Let $\mathfrak{p} \in \text{Proj } S$. Define

$$S_{(\mathfrak{p})}$$

to be the degree-zero part of the homogeneous localization of S at $\mathfrak{p}$.

Concretely,

$$S_{(\mathfrak{p})} = \left\{ \frac{a}{b} : a, b \in S \text{ homogeneous, } \deg a = \deg b, b \notin \mathfrak{p} \right\}.$$

The condition

$$\deg a = \deg b$$

ensures that the fraction

$$\frac{a}{b}$$

has degree zero.

5.8 Why Degree Zero?

The degree-zero condition is the algebraic form of projective invariance.

A regular function on projective space should be independent of the chosen representative of a projective point.

If

$$(a_0, \dots, a_n) \sim (\lambda a_0, \dots, \lambda a_n),$$

then a well-defined projective function should have the same value on both representatives.

Let $F, G \in k[x_0, \dots, x_n]$ be homogeneous polynomials of the same degree d . Then

$$F(\lambda a) = \lambda^d F(a)$$

and

$$G(\lambda a) = \lambda^d G(a).$$

Therefore

$$\frac{F(\lambda a)}{G(\lambda a)} = \frac{\lambda^d F(a)}{\lambda^d G(a)} = \frac{F(a)}{G(a)}.$$

Thus ratios of homogeneous polynomials of the same degree are invariant under scalar multiplication.

This is why regular functions on projective schemes are locally represented by degree-zero homogeneous fractions.

5.9 The Local Ring $S_{(\mathfrak{p})}$

Proposition 5.4. *For every $\mathfrak{p} \in \text{Proj } S$, the ring $S_{(\mathfrak{p})}$ is local. Its maximal ideal is*

$$\mathfrak{p}_{(\mathfrak{p})} = \left\{ \frac{a}{b} \in S_{(\mathfrak{p})} : a \in \mathfrak{p} \right\}.$$

Proof. Let

$$\frac{a}{b} \in S_{(\mathfrak{p})}$$

with

$$a, b$$

homogeneous of the same degree and

$$b \notin \mathfrak{p}.$$

Suppose

$$\frac{a}{b} \notin \mathfrak{p}_{(\mathfrak{p})}.$$

Then

$$a \notin \mathfrak{p}.$$

Since also

$$b \notin \mathfrak{p},$$

the fraction

$$\frac{b}{a}$$

is defined in $S_{(\mathfrak{p})}$. Moreover,

$$\frac{a}{b} \cdot \frac{b}{a} = 1.$$

Thus every element outside $\mathfrak{p}_{(\mathfrak{p})}$ is invertible.

Therefore $S_{(\mathfrak{p})}$ is a local ring, with maximal ideal

$$\mathfrak{p}_{(\mathfrak{p})}.$$

□

6 The Structure Sheaf on $\text{Proj } S$

The structure sheaf on $\text{Proj } S$ is defined using local degree-zero fractions.

Let

$$U \subseteq \text{Proj } S$$

be open.

A section

$$s \in \mathcal{O}_{\text{Proj } S}(U)$$

is a function

$$s : U \longrightarrow \prod_{\mathfrak{p} \in U} S_{(\mathfrak{p})}$$

such that

$$s(\mathfrak{p}) \in S_{(\mathfrak{p})}$$

for every

$$\mathfrak{p} \in U,$$

and such that s is locally represented by a homogeneous fraction of degree zero.

More explicitly, for every point

$$\mathfrak{p} \in U,$$

there exists an open neighbourhood

$$V \subseteq U$$

of $\mathfrak{p}$, and homogeneous elements

$$a, b \in S$$

of the same degree, such that

$$b \notin \mathfrak{q}$$

for every

$$\mathfrak{q} \in V,$$

and

$$s(\mathfrak{q}) = \frac{a}{b}$$

in $S_{(\mathfrak{q})}$ for all

$$\mathfrak{q} \in V.$$

Thus locally regular functions on $\text{Proj } S$ are exactly degree-zero homogeneous fractions.

6.1 The Stalks of the Structure Sheaf

The projective analogue of the affine identity

$$\mathcal{O}_{\text{Spec } A, \mathfrak{p}} \cong A_{\mathfrak{p}}$$

is the following.

Theorem 6.1. *For every point*

$$\mathfrak{q} \in \text{Proj } S,$$

there is a natural isomorphism

$$\mathcal{O}_{\text{Proj } S, \mathfrak{q}} \cong S_{(\mathfrak{q})}.$$

Proof. Define

$$\Phi : \mathcal{O}_{\text{Proj } S, \mathfrak{q}} \longrightarrow S_{(\mathfrak{q})}$$

by evaluation at $\mathfrak{q}$.

If a germ is represented by a section

$$s \in \mathcal{O}_{\text{Proj } S}(U), \quad \mathfrak{q} \in U,$$

then define

$$\Phi(s_{\mathfrak{q}}) = s(\mathfrak{q}).$$

This is a well-defined ring homomorphism.

We show that Φ is surjective.

Let

$$\frac{a}{b} \in S_{(\mathfrak{q})}.$$

Then $a, b \in S$ are homogeneous of the same degree and

$$b \notin \mathfrak{q}.$$

Therefore

$$\mathfrak{q} \in D_+(b).$$

Define a section on $D_+(b)$ by

$$s(\mathfrak{p}) = \frac{a}{b}.$$

This is a regular section by definition. Its germ at $\mathfrak{q}$ maps to a/b . Thus Φ is surjective.

Now we prove injectivity.

Suppose

$$\Phi(s_{\mathfrak{q}}) = 0.$$

Since s is locally represented by a homogeneous fraction, after shrinking around $\mathfrak{q}$, we may assume that

$$s(\mathfrak{p}) = \frac{a}{b}$$

on an open neighbourhood V of $\mathfrak{q}$, where a, b are homogeneous of the same degree and

$$b \notin \mathfrak{p}$$

for every $\mathfrak{p} \in V$.

The equality

$$s(\mathfrak{q}) = 0$$

means that

$$\frac{a}{b} = 0$$

in $S_{(\mathfrak{q})}$.

Therefore there exists a homogeneous element

$$h \notin \mathfrak{q}$$

such that

$$ha = 0.$$

Now restrict to the smaller neighbourhood

$$V \cap D_+(h).$$

On this neighbourhood,

$$h \notin \mathfrak{p}.$$

Since

$$ha = 0,$$

we get

$$\frac{a}{b} = 0$$

in $S_{(\mathfrak{p})}$ for all $\mathfrak{p} \in V \cap D_+(h)$.

Hence s is zero on a neighbourhood of $\mathfrak{q}$, so the germ $s_{\mathfrak{q}}$ is zero.

Thus Φ is injective.

Therefore

$$\mathcal{O}_{\text{Proj } S, \mathfrak{q}} \cong S_{(\mathfrak{q})}.$$

□

6.2 Affine Charts of $\text{Proj } S$

Let $f \in S_+$ be homogeneous, and set

$$d = \deg f.$$

The localization S_f is naturally graded. Define

$$S_{(f)} = (S_f)_0.$$

Thus $S_{(f)}$ is the degree-zero part of S_f .

Concretely,

$$S_{(f)} = \left\{ \frac{a}{f^m} : a \in S_{md} \right\}.$$

The key local theorem is:

Theorem 6.2. *For every homogeneous $f \in S_+$, there is an isomorphism of locally ringed spaces*

$$(D_+(f), \mathcal{O}_{\text{Proj } S}|_{D_+(f)}) \cong \text{Spec } S_{(f)}.$$

This theorem is the projective analogue of the affine identity

$$D(f) \cong \text{Spec } A_f.$$

In the projective case, however, the ring is not the whole localization S_f , but only its degree-zero part:

$$S_{(f)} = (S_f)_0.$$

6.3 The Map $D_+(f) \rightarrow \text{Spec } S_{(f)}$

Let

$$\mathfrak{p} \in D_+(f).$$

Then

$$f \notin \mathfrak{p}.$$

Therefore we can localize $\mathfrak{p}$ in S_f .

Define

$$\psi(\mathfrak{p}) = \mathfrak{p}S_f \cap S_{(f)}.$$

This is a prime ideal of $S_{(f)}$, because $\mathfrak{p}S_f$ is a prime ideal of S_f , and the contraction of a prime ideal is prime.

Thus we obtain a map

$$\psi : D_+(f) \longrightarrow \text{Spec } S_{(f)}.$$

6.4 The Inverse Map

The inverse construction is more subtle.

Let

$$\mathfrak{q} \in \text{Spec } S_{(f)}.$$

For each $n \geq 0$, define

$$\mathfrak{q}_n = \left\{ s \in S_n : \frac{s^d}{f^n} \in \mathfrak{q} \right\},$$

where

$$d = \deg f.$$

This definition makes sense because if

$$s \in S_n,$$

then

$$s^d \in S_{dn}$$

and

$$f^n \in S_{dn}.$$

Thus

$$\frac{s^d}{f^n}$$

has degree zero, so it lies in $S_{(f)}$.

Now define

$$\theta(\mathfrak{q}) = \bigoplus_{n \geq 0} \mathfrak{q}_n.$$

Lemma 6.3. *Each $\mathfrak{q}_n$ is closed under addition.*

Proof. Let

$$s, t \in \mathfrak{q}_n.$$

Then

$$\frac{s^d}{f^n} \in \mathfrak{q}$$

and

$$\frac{t^d}{f^n} \in \mathfrak{q}.$$

We must show that

$$s + t \in \mathfrak{q}_n.$$

That is, we must show

$$\frac{(s + t)^d}{f^n} \in \mathfrak{q}.$$

By the binomial theorem,

$$\frac{(s + t)^d}{f^n} = \sum_{i=0}^d \binom{d}{i} \frac{s^i t^{d-i}}{f^n}.$$

Each summand has degree zero.

Now observe that

$$\left(\frac{s^i t^{d-i}}{f^n} \right)^d = \left(\frac{s^d}{f^n} \right)^i \left(\frac{t^d}{f^n} \right)^{d-i}.$$

Since

$$\frac{s^d}{f^n}, \frac{t^d}{f^n} \in \mathfrak{q},$$

the right-hand side lies in $\mathfrak{q}$.

Because $\mathfrak{q}$ is prime, it is radical. Therefore

$$\frac{s^i t^{d-i}}{f^n} \in \mathfrak{q}.$$

Hence every term in the binomial expansion lies in $\mathfrak{q}$. Therefore

$$\frac{(s + t)^d}{f^n} \in \mathfrak{q}.$$

Thus

$$s + t \in \mathfrak{q}_n.$$

□

Lemma 6.4. *The set*

$$\theta(\mathfrak{q}) = \bigoplus_{n \geq 0} \mathfrak{q}_n$$

is a homogeneous ideal of S .

Proof. By construction,

$$\mathfrak{q}_n \subseteq S_n.$$

Therefore

$$\theta(\mathfrak{q})$$

is homogeneous.

It remains to show that it is an ideal.

Let

$$s \in \mathfrak{q}_n$$

and let

$$r \in S_m$$

be homogeneous.

Then

$$rs \in S_{m+n}.$$

We compute:

$$\frac{(rs)^d}{f^{m+n}} = \frac{r^d}{f^m} \cdot \frac{s^d}{f^n}.$$

The first factor

$$\frac{r^d}{f^m}$$

has degree zero, so it belongs to $S_{(f)}$.

The second factor belongs to $\mathfrak{q}$, because $s \in \mathfrak{q}_n$.

Therefore

$$\frac{(rs)^d}{f^{m+n}} \in \mathfrak{q}.$$

Hence

$$rs \in \mathfrak{q}_{m+n}.$$

Therefore $\theta(\mathfrak{q})$ is a homogeneous ideal of S . □

Lemma 6.5. *The homogeneous ideal*

$$\theta(\mathfrak{q})$$

is prime.

Proof. Let

$$a \in S_m, \quad b \in S_n$$

be homogeneous elements, and suppose

$$ab \in \theta(\mathfrak{q}).$$

Then

$$ab \in \mathfrak{q}_{m+n}.$$

Therefore

$$\frac{(ab)^d}{f^{m+n}} \in \mathfrak{q}.$$

But

$$\frac{(ab)^d}{f^{m+n}} = \frac{a^d}{f^m} \cdot \frac{b^d}{f^n}.$$

Since $\mathfrak{q}$ is prime, either

$$\frac{a^d}{f^m} \in \mathfrak{q}$$

or

$$\frac{b^d}{f^n} \in \mathfrak{q}.$$

Thus either

$$a \in \mathfrak{q}_m$$

or

$$b \in \mathfrak{q}_n.$$

Therefore either

$$a \in \theta(\mathfrak{q})$$

or

$$b \in \theta(\mathfrak{q}).$$

Hence $\theta(\mathfrak{q})$ is prime. $\square$

Lemma 6.6. *The prime ideal $\theta(\mathfrak{q})$ belongs to $D_+(f)$.*

Proof. We must show that

$$f \notin \theta(\mathfrak{q}).$$

Since

$$f \in S_d,$$

we would have

$$f \in \mathfrak{q}_d$$

only if

$$\frac{f^d}{f^d} \in \mathfrak{q}.$$

But

$$\frac{f^d}{f^d} = 1,$$

and no prime ideal contains 1.

Therefore

$$f \notin \theta(\mathfrak{q}).$$

Hence

$$\theta(\mathfrak{q}) \in D_+(f).$$

□

Thus we obtain a map

$$\theta : \operatorname{Spec} S_{(f)} \longrightarrow D_+(f).$$

6.5 The Maps Are Inverse

We now show that

$$\psi \circ \theta = \operatorname{id}$$

and

$$\theta \circ \psi = \operatorname{id}.$$

Let

$$\mathfrak{q} \in \operatorname{Spec} S_{(f)}.$$

An arbitrary element of $S_{(f)}$ has the form

$$\frac{a}{f^m},$$

where

$$a \in S_{md}.$$

Then

$$\frac{a}{f^m} \in \psi(\theta(\mathfrak{q}))$$

if and only if

$$a \in \theta(\mathfrak{q}).$$

By definition of θ , this is equivalent to

$$\frac{a^d}{f^{md}} \in \mathfrak{q}.$$

But

$$\frac{a^d}{f^{md}} = \left(\frac{a}{f^m} \right)^d.$$

Since $\mathfrak{q}$ is prime, hence radical, this is equivalent to

$$\frac{a}{f^m} \in \mathfrak{q}.$$

Therefore

$$\psi(\theta(\mathfrak{q})) = \mathfrak{q}.$$

Conversely, let

$$\mathfrak{p} \in D_+(f).$$

Let

$$s \in S_n$$

be homogeneous.

Then

$$s \in \theta(\psi(\mathfrak{p}))$$

if and only if

$$\frac{s^d}{f^n} \in \psi(\mathfrak{p}).$$

By definition of ψ , this is equivalent to

$$\frac{s^d}{f^n} \in \mathfrak{p}S_f.$$

Since

$$f \notin \mathfrak{p},$$

the denominator f^n is invertible after localization. Therefore this is equivalent to

$$s^d \in \mathfrak{p}.$$

Because $\mathfrak{p}$ is prime, this is equivalent to

$$s \in \mathfrak{p}.$$

Thus every homogeneous element belongs to $\theta(\psi(\mathfrak{p}))$ if and only if it belongs to $\mathfrak{p}$.

Since both ideals are homogeneous,

$$\theta(\psi(\mathfrak{p})) = \mathfrak{p}.$$

Therefore

$$D_+(f) \cong \operatorname{Spec} S_{(f)}$$

as topological spaces.

6.6 Compatibility with the Structure Sheaves

The homeomorphism

$$D_+(f) \cong \operatorname{Spec} S_{(f)}$$

also identifies the structure sheaves.

On $D_+(f)$, sections of $\mathcal{O}_{\text{Proj } S}$ are locally represented by fractions

$$\frac{a}{b},$$

where $a, b \in S$ are homogeneous of the same degree.

On $\text{Spec } S_{(f)}$, sections of the affine structure sheaf are locally represented by fractions in localizations of $S_{(f)}$.

These descriptions agree.

More concretely, if $g \in S$ is homogeneous, then

$$D_+(f) \cap D_+(g)$$

corresponds to the affine principal open subset

$$D\left(\frac{g^d}{f^{\deg g}}\right) \subseteq \text{Spec } S_{(f)}.$$

Indeed, for

$$\mathfrak{p} \in D_+(f),$$

we have

$$g \notin \mathfrak{p}$$

if and only if

$$\frac{g^d}{f^{\deg g}} \notin \psi(\mathfrak{p}).$$

On such basic opens, both structure sheaves are described by the same degree-zero fractions.

Therefore there is an isomorphism of locally ringed spaces

$$(D_+(f), \mathcal{O}_{\text{Proj } S}|_{D_+(f)}) \cong \text{Spec } S_{(f)}.$$

6.7 Proj S is a Scheme

Theorem 6.7. *For every graded ring*

$$S = \bigoplus_{d \geq 0} S_d,$$

the locally ringed space $\text{Proj } S$ is a scheme.

Proof. The open subsets

$$D_+(f), \quad f \in S_+ \text{ homogeneous},$$

cover $\text{Proj } S$.

For each such f , we have shown that

$$D_+(f) \cong \text{Spec } S_{(f)}$$

as locally ringed spaces.

Therefore $\text{Proj } S$ is covered by affine schemes.

Hence

$$\text{Proj } S$$

is a scheme. $\square$

6.8 Projective Space as a Proj

Let

$$S = k[x_0, \dots, x_n]$$

with the standard grading.

Then

$$\text{Proj } k[x_0, \dots, x_n]$$

is projective n -space over k , denoted

$$\mathbb{P}_k^n.$$

Thus

$$\mathbb{P}_k^n = \text{Proj } k[x_0, \dots, x_n].$$

6.9 Affine Charts of Projective Space

Consider

$$\mathbb{P}_k^n = \text{Proj } k[x_0, \dots, x_n].$$

For each i , the open subset

$$D_+(x_i)$$

is the set of projective points where

$$x_i \neq 0.$$

Since $x_i \neq 0$, we may divide all coordinates by x_i .

The coordinate ring of this affine chart is

$$k \left[\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i} \right].$$

Therefore

$$D_+(x_i) \cong \text{Spec } k \left[\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i} \right] \cong \mathbb{A}_k^n.$$

Thus projective space is covered by $n + 1$ affine charts:

$$\mathbb{P}_k^n = D_+(x_0) \cup D_+(x_1) \cup \dots \cup D_+(x_n).$$

Each chart is affine n -space.

6.10 Example: The Projective Line

Let

$$S = k[x_0, x_1].$$

Then

$$\text{Proj } S = \mathbb{P}_k^1.$$

There are two standard affine charts:

$$D_+(x_0)$$

and

$$D_+(x_1).$$

On $D_+(x_0)$, define

$$t = \frac{x_1}{x_0}.$$

Then

$$D_+(x_0) \cong \text{Spec } k[t].$$

On $D_+(x_1)$, define

$$u = \frac{x_0}{x_1}.$$

Then

$$D_+(x_1) \cong \text{Spec } k[u].$$

On the overlap

$$D_+(x_0) \cap D_+(x_1),$$

we have

$$u = \frac{1}{t}.$$

Thus $\mathbb{P}_k^1$ is obtained by gluing two affine lines

$$\mathbb{A}_k^1$$

along the open subsets where $t \neq 0$ and $u \neq 0$, using the transition rule

$$u = t^{-1}.$$

6.11 Example: The Projective Plane

Let

$$S = k[x_0, x_1, x_2].$$

Then

$$\text{Proj } S = \mathbb{P}_k^2.$$

The standard affine charts are

$$D_+(x_0), \quad D_+(x_1), \quad D_+(x_2).$$

On $D_+(x_0)$,

$$D_+(x_0) \cong \operatorname{Spec} k \left[\frac{x_1}{x_0}, \frac{x_2}{x_0} \right] \cong \mathbb{A}_k^2.$$

On $D_+(x_1)$,

$$D_+(x_1) \cong \operatorname{Spec} k \left[\frac{x_0}{x_1}, \frac{x_2}{x_1} \right] \cong \mathbb{A}_k^2.$$

On $D_+(x_2)$,

$$D_+(x_2) \cong \operatorname{Spec} k \left[\frac{x_0}{x_2}, \frac{x_1}{x_2} \right] \cong \mathbb{A}_k^2.$$

Thus projective 2-space is obtained by gluing three affine planes.

6.12 Closed Points of Projective Space

Assume now that k is algebraically closed.

The closed points of

$$\operatorname{Proj} k[x_0, \dots, x_n]$$

are exactly the usual projective points

$$[a_0 : \dots : a_n].$$

Equivalently,

$$|\operatorname{Proj} k[x_0, \dots, x_n]|_{\text{closed}} \cong (k^{n+1} \setminus \{0\})/k^*.$$

The equivalence relation is

$$(a_0, \dots, a_n) \sim (\lambda a_0, \dots, \lambda a_n), \quad \lambda \in k^*.$$

A point

$$[a_0 : \dots : a_n]$$

corresponds to the homogeneous prime ideal

$$\mathfrak{p}_a = \{F \in k[x_0, \dots, x_n] \text{ homogeneous} : F(a_0, \dots, a_n) = 0\}.$$

This ideal depends only on the projective class.

Indeed, if

$$b = \lambda a, \quad \lambda \in k^*,$$

and F is homogeneous of degree d , then

$$F(b) = F(\lambda a) = \lambda^d F(a).$$

Therefore

$$F(a) = 0$$

if and only if

$$F(b) = 0.$$

Thus

$$\mathfrak{p}_a = \mathfrak{p}_b.$$

Conversely, every closed point of projective space lies in some affine chart

$$D_+(x_i) \cong \mathbb{A}_k^n.$$

Since k is algebraically closed, the closed points of affine n -space are the usual k -rational points. Therefore every closed point of projective space comes from a projective point

$$[a_0 : \cdots : a_n].$$

Hence

$$|\mathrm{Proj} k[x_0, \dots, x_n]|_{\mathrm{closed}} \cong (k^{n+1} \setminus \{0\})/k^*.$$

6.13 Closed Points of Projective Varieties

Let

$$I \subseteq k[x_0, \dots, x_n]$$

be a homogeneous ideal.

Then

$$\mathrm{Proj} k[x_0, \dots, x_n]/I$$

is the projective scheme cut out by the homogeneous equations in I .

Its closed points are precisely the projective points

$$[a_0 : \cdots : a_n] \in \mathbb{P}_k^n$$

such that

$$F(a_0, \dots, a_n) = 0$$

for every homogeneous polynomial

$$F \in I.$$

Thus

$$|\mathrm{Proj} k[x_0, \dots, x_n]/I|_{\mathrm{closed}} \cong \{[a_0 : \cdots : a_n] \in \mathbb{P}_k^n : F(a_0, \dots, a_n) = 0 \text{ for all homogeneous } F \in I\}.$$

Equivalently, the closed points of

$$\mathrm{Proj} k[x_0, \dots, x_n]/I$$

are projective equivalence classes of nonzero points

$$(a_0, \dots, a_n) \in k^{n+1} \setminus \{0\}$$

satisfying all homogeneous equations in I .

6.14 Conceptual Summary

The construction of $\text{Proj } S$ does three things.

First, it keeps only homogeneous data, because projective geometry is invariant under scaling.

Second, it removes the irrelevant ideal S_+ , because S_+ corresponds to the forbidden origin.

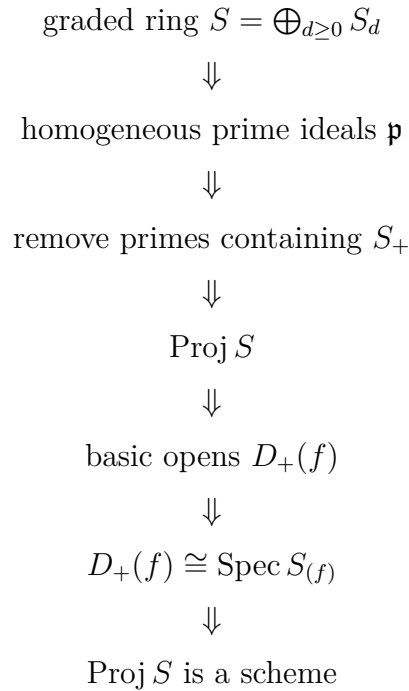
Third, it defines local affine charts by inverting homogeneous elements and taking degree-zero fractions:

$$D_+(f) \cong \text{Spec } S_{(f)}.$$

Therefore

$\text{Proj } S =$ the scheme obtained by gluing the affine charts $\text{Spec } S_{(f)}$ for homogeneous $f \in S_+$.

6.15 Conceptual Diagram



6.16 Final Dictionary

Affine construction	Projective construction
A is a ring	$S = \bigoplus_{d \geq 0} S_d$ is a graded ring
$\operatorname{Spec} A$	$\operatorname{Proj} S$
prime ideals	homogeneous prime ideals not containing S_+
$V(I)$	$V_+(I)$
$D(f)$	$D_+(f)$
A_f	$S_{(f)} = (S_f)_0$
$D(f) \cong \operatorname{Spec} A_f$	$D_+(f) \cong \operatorname{Spec} S_{(f)}$
$A_{\mathfrak{p}}$	$S_{(\mathfrak{p})}$
$\mathcal{O}_{\operatorname{Spec} A, \mathfrak{p}} \cong A_{\mathfrak{p}}$	$\mathcal{O}_{\operatorname{Proj} S, \mathfrak{p}} \cong S_{(\mathfrak{p})}$

The most important local formula is

$$D_+(f) \cong \operatorname{Spec} S_{(f)}.$$

The most important conceptual idea is

$$\text{projective regular functions are degree-zero homogeneous fractions.}$$

7 Examples of the Construction of $\operatorname{Proj} S$

In this section we give examples illustrating the construction of

$$\operatorname{Proj} S$$

for graded rings. The main ideas are:

$$\operatorname{Proj} S = \{\text{homogeneous prime ideals of } S \text{ not containing } S_+\},$$

where

$$S_+ = \bigoplus_{d \geq 1} S_d$$

is the irrelevant ideal.

The most important local fact is

$$D_+(f) \cong \operatorname{Spec} S_{(f)},$$

where

$$S_{(f)} = (S_f)_0$$

is the degree-zero part of the localization S_f .

Thus, in examples, one usually studies $\operatorname{Proj} S$ by looking at affine charts

$$D_+(f).$$

7.1 The Basic Example: $\text{Proj } k[x_0, \dots, x_n]$

Let

$$S = k[x_0, \dots, x_n]$$

with the standard grading:

$$S_d = \{\text{homogeneous polynomials of degree } d\}.$$

Then

$$S_+ = (x_0, \dots, x_n).$$

By definition,

$$\text{Proj } S$$

is the set of homogeneous prime ideals of S not containing S_+ .

The fundamental example is

$$\boxed{\text{Proj } k[x_0, \dots, x_n] = \mathbb{P}_k^n.}$$

Thus ordinary projective n -space is the Proj of the standard graded polynomial ring.

The basic affine charts are

$$D_+(x_i) = \{\mathfrak{p} \in \text{Proj } S : x_i \notin \mathfrak{p}\}.$$

Each chart is affine n -space:

$$D_+(x_i) \cong \text{Spec } k \left[\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i} \right] \cong \mathbb{A}_k^n.$$

Therefore

$$\mathbb{P}_k^n = D_+(x_0) \cup D_+(x_1) \cup \dots \cup D_+(x_n),$$

and each $D_+(x_i)$ is an affine chart.

7.2 Example: $\mathbb{P}_k^1$

Let

$$S = k[x_0, x_1].$$

Then

$$\text{Proj } S = \mathbb{P}_k^1.$$

There are two standard affine charts:

$$D_+(x_0) \quad \text{and} \quad D_+(x_1).$$

On $D_+(x_0)$, we invert x_0 . The degree-zero part of S_{x_0} is

$$S_{(x_0)} = k \left[\frac{x_1}{x_0} \right].$$

Set

$$t = \frac{x_1}{x_0}.$$

Then

$$D_+(x_0) \cong \operatorname{Spec} k[t] \cong \mathbb{A}_k^1.$$

On the other chart,

$$S_{(x_1)} = k \left[\frac{x_0}{x_1} \right].$$

Set

$$u = \frac{x_0}{x_1}.$$

Then

$$D_+(x_1) \cong \operatorname{Spec} k[u] \cong \mathbb{A}_k^1.$$

On the overlap,

$$D_+(x_0) \cap D_+(x_1),$$

we have

$$u = \frac{x_0}{x_1} = \left(\frac{x_1}{x_0} \right)^{-1} = t^{-1}.$$

So

$$\mathbb{P}_k^1$$

is obtained by gluing two affine lines along the open subsets

$$t \neq 0$$

and

$$u \neq 0$$

using the transition rule

$$u = t^{-1}.$$

Geometrically,

$$\boxed{\mathbb{P}_k^1 = \mathbb{A}_k^1 \cup \{\infty\}.$$

7.3 Example: Closed Points of $\mathbb{P}_k^1$

Assume k is algebraically closed.

A closed point of

$$\mathbb{P}_k^1 = \operatorname{Proj} k[x_0, x_1]$$

is a projective equivalence class

$$[a_0 : a_1],$$

where

$$(a_0, a_1) \neq (0, 0),$$

and

$$[a_0 : a_1] = [\lambda a_0 : \lambda a_1]$$

for all

$$\lambda \in k^*.$$

If

$$a_0 \neq 0,$$

then

$$[a_0 : a_1] = [1 : a_1/a_0].$$

These points lie in the affine chart $D_+(x_0)$.

If

$$a_0 = 0,$$

then since the point is nonzero, we must have

$$a_1 \neq 0.$$

Hence

$$[0 : a_1] = [0 : 1].$$

This is the point at infinity.

Therefore

$$\mathbb{P}_k^1(k) = \{[1 : t] : t \in k\} \cup \{[0 : 1]\}.$$

So

$$\boxed{\mathbb{P}_k^1(k) = k \cup \{\infty\}.$$

7.4 Example: $\mathbb{P}_k^2$

Let

$$S = k[x_0, x_1, x_2].$$

Then

$$\text{Proj } S = \mathbb{P}_k^2.$$

There are three standard affine charts:

$$D_+(x_0), \quad D_+(x_1), \quad D_+(x_2).$$

On $D_+(x_0)$,

$$S_{(x_0)} = k\left[\frac{x_1}{x_0}, \frac{x_2}{x_0}\right].$$

So

$$D_+(x_0) \cong \text{Spec } k\left[\frac{x_1}{x_0}, \frac{x_2}{x_0}\right] \cong \mathbb{A}_k^2.$$

Similarly,

$$D_+(x_1) \cong \operatorname{Spec} k \left[\frac{x_0}{x_1}, \frac{x_2}{x_1} \right] \cong \mathbb{A}_k^2,$$

and

$$D_+(x_2) \cong \operatorname{Spec} k \left[\frac{x_0}{x_2}, \frac{x_1}{x_2} \right] \cong \mathbb{A}_k^2.$$

Thus

$$\boxed{\mathbb{P}_k^2 = D_+(x_0) \cup D_+(x_1) \cup D_+(x_2),}$$

where each chart is an affine plane.

7.5 Example: A Projective Line Inside $\mathbb{P}_k^2$

Let

$$S = k[x_0, x_1, x_2],$$

and consider the homogeneous ideal

$$I = (x_2).$$

Then

$$\operatorname{Proj} S/I = \operatorname{Proj} k[x_0, x_1, x_2]/(x_2).$$

But

$$k[x_0, x_1, x_2]/(x_2) \cong k[x_0, x_1].$$

Therefore

$$\operatorname{Proj} S/I \cong \operatorname{Proj} k[x_0, x_1] = \mathbb{P}_k^1.$$

Geometrically, the equation

$$x_2 = 0$$

defines a projective line in $\mathbb{P}_k^2$.

So

$$\boxed{V_+(x_2) \subseteq \mathbb{P}_k^2 \text{ is a projective line.}}$$

Its closed points are

$$[a_0 : a_1 : 0].$$

7.6 Example: A Projective Conic

Let

$$S = k[x_0, x_1, x_2],$$

and consider the homogeneous polynomial

$$F = x_0x_2 - x_1^2.$$

Let

$$I = (F).$$

Then

$$X = \text{Proj } S/I$$

is the projective conic

$$x_0x_2 = x_1^2$$

inside

$$\mathbb{P}_k^2.$$

Its closed points are projective classes

$$[a_0 : a_1 : a_2] \in \mathbb{P}_k^2$$

such that

$$a_0a_2 - a_1^2 = 0.$$

Therefore

$$X(k) = \{[a_0 : a_1 : a_2] \in \mathbb{P}_k^2 : a_0a_2 = a_1^2\}.$$

Now look at the affine chart $D_+(x_0)$. There we set

$$u = \frac{x_1}{x_0}, \quad v = \frac{x_2}{x_0}.$$

Divide the equation

$$x_0x_2 = x_1^2$$

by x_0^2 . We get

$$\frac{x_2}{x_0} = \left(\frac{x_1}{x_0}\right)^2.$$

Thus on this chart,

$$v = u^2.$$

Therefore the conic locally looks like the affine parabola

$$v = u^2.$$

This shows how a homogeneous projective equation becomes an affine equation on a chart.

7.7 Example: A Projective Hypersurface

Let

$$S = k[x_0, \dots, x_n],$$

and let

$$F \in S_d$$

be homogeneous of degree d .

Then

$$X = \text{Proj } S/(F)$$

is the projective hypersurface

$$F = 0$$

inside

$$\mathbb{P}_k^n.$$

The closed points of X are

$$[a_0 : \cdots : a_n] \in \mathbb{P}_k^n$$

such that

$$F(a_0, \dots, a_n) = 0.$$

So

$$X(k) = \{[a_0 : \cdots : a_n] \in \mathbb{P}_k^n : F(a_0, \dots, a_n) = 0\}.$$

For example, if

$$F = x_0^2 + x_1^2 + x_2^2,$$

then

$$\text{Proj } k[x_0, x_1, x_2]/(F)$$

is a projective conic.

If

$$F = x_0^3 + x_1^3 + x_2^3,$$

then

$$\text{Proj } k[x_0, x_1, x_2]/(F)$$

is a plane cubic curve.

7.8 Example: Why Non-Homogeneous Equations Do Not Work

Consider

$$f = x_0 + 1 \in k[x_0, x_1].$$

This is not homogeneous.

If we try to define a projective equation

$$x_0 + 1 = 0,$$

it does not make sense on projective points.

Suppose

$$(a_0, a_1)$$

satisfies

$$a_0 + 1 = 0.$$

Then

$$a_0 = -1.$$

But the same projective point can be represented by

$$(\lambda a_0, \lambda a_1).$$

Then the equation becomes

$$\lambda a_0 + 1 = 0.$$

This condition is not equivalent to

$$a_0 + 1 = 0.$$

So the vanishing condition is not invariant under scaling.

That is why projective closed subsets must be defined by homogeneous equations.

By contrast, if

$$F = x_0^2 + x_1^2$$

is homogeneous, then

$$F(\lambda a_0, \lambda a_1) = \lambda^2 F(a_0, a_1).$$

Therefore

$$F(a_0, a_1) = 0$$

if and only if

$$F(\lambda a_0, \lambda a_1) = 0.$$

So homogeneous equations are well-defined on projective space.

7.9 Example: Degree-Zero Fractions as Functions

Let

$$S = k[x_0, x_1].$$

On

$$\mathbb{P}_k^1 = \text{Proj } S,$$

the expression

$$\frac{x_1}{x_0}$$

is a valid regular function on the chart

$$D_+(x_0).$$

Indeed, x_1 and x_0 are homogeneous of the same degree:

$$\deg x_1 = \deg x_0 = 1.$$

Hence

$$\frac{x_1}{x_0}$$

has degree zero.

It is also invariant under scaling:

$$\frac{\lambda x_1}{\lambda x_0} = \frac{x_1}{x_0}.$$

However,

$$\frac{x_1^2}{x_0}$$

is not degree zero, because

$$\deg x_1^2 = 2, \quad \deg x_0 = 1.$$

So

$$\frac{x_1^2}{x_0}$$

is not a regular function on projective space.

The corresponding degree-zero expression would be

$$\frac{x_1^2}{x_0^2}.$$

Thus projective regular functions are locally represented by fractions

$$\frac{F}{G}$$

where

$$\deg F = \deg G.$$

7.10 Example: Computing $S_{(x_0)}$

Let

$$S = k[x_0, x_1, x_2].$$

We compute

$$S_{(x_0)}.$$

The localization is

$$S_{x_0} = k[x_0, x_1, x_2, x_0^{-1}].$$

The degree-zero part consists of fractions

$$\frac{F}{x_0^m},$$

where F is homogeneous of degree m .

Every such fraction can be written in terms of

$$\frac{x_1}{x_0} \quad \text{and} \quad \frac{x_2}{x_0}.$$

Indeed, if

$$F = x_1^a x_2^b x_0^c$$

is a monomial of degree m , then

$$a + b + c = m.$$

Therefore

$$\frac{x_1^a x_2^b x_0^c}{x_0^m} = \left(\frac{x_1}{x_0}\right)^a \left(\frac{x_2}{x_0}\right)^b.$$

Hence

$$S_{(x_0)} = k \left[\frac{x_1}{x_0}, \frac{x_2}{x_0} \right].$$

Therefore

$$D_+(x_0) \cong \operatorname{Spec} k \left[\frac{x_1}{x_0}, \frac{x_2}{x_0} \right].$$

7.11 Example: $\operatorname{Proj} k[x]$ is One Point

Let

$$S = k[x],$$

with the standard grading

$$\deg x = 1.$$

Then

$$S_+ = (x).$$

The homogeneous prime ideals of $k[x]$ are

$$(0) \quad \text{and} \quad (x).$$

The ideal

$$(x)$$

contains S_+ , so it is excluded from $\operatorname{Proj} S$.

The ideal

$$(0)$$

does not contain S_+ , so it remains.

Hence

$$\operatorname{Proj} k[x]$$

has one point.

Indeed,

$$\operatorname{Proj} k[x] \cong \operatorname{Spec} k.$$

The affine chart is

$$D_+(x) = \operatorname{Proj} k[x],$$

and

$$S_{(x)} = (k[x]_x)_0 = k.$$

Therefore

$$D_+(x) \cong \operatorname{Spec} k.$$

Thus

$$\operatorname{Proj} k[x] \cong \operatorname{Spec} k = \mathbb{P}_k^0.$$

7.12 Example: $\operatorname{Proj} k$ is Empty

Let

$$S = k$$

with grading

$$S_0 = k, \quad S_d = 0 \quad \text{for } d > 0.$$

Then

$$S_+ = 0.$$

A point of $\operatorname{Proj} S$ is a homogeneous prime ideal $\mathfrak{p}$ such that

$$S_+ \not\subseteq \mathfrak{p}.$$

But

$$S_+ = 0$$

is contained in every prime ideal.

Therefore there are no points.

Hence

$$\operatorname{Proj} k = \emptyset.$$

This example shows why the irrelevant ideal condition matters.

7.13 Example: Quotient by the Irrelevant Ideal Gives Empty Proj

Let

$$S = k[x_0, \dots, x_n].$$

Consider the quotient

$$S/S_+ = k[x_0, \dots, x_n]/(x_0, \dots, x_n) \cong k.$$

Then

$$\operatorname{Proj}(S/S_+) = \operatorname{Proj} k = \emptyset.$$

So killing all positive-degree elements destroys projective geometry.

Geometrically, this corresponds to keeping only the forbidden origin, which is removed in projective space.

Thus

$$\operatorname{Proj}(S/S_+) = \emptyset.$$

7.14 Example: A Non-Reduced Projective Scheme

Let

$$S = k[x_0, x_1],$$

and let

$$I = (x_1^2).$$

Then

$$X = \text{Proj } S/I = \text{Proj } k[x_0, x_1]/(x_1^2).$$

The underlying closed subset is defined by

$$x_1^2 = 0.$$

Set-theoretically, this is the same as

$$x_1 = 0.$$

Thus the underlying topological space is one point:

$$[1 : 0].$$

But scheme-theoretically, X is not reduced.

On the chart

$$D_+(x_0),$$

set

$$t = \frac{x_1}{x_0}.$$

The equation

$$x_1^2 = 0$$

becomes

$$t^2 = 0.$$

So the affine chart is

$$\text{Spec } k[t]/(t^2).$$

Therefore X is a doubled point in projective space.

This example shows that Proj remembers scheme structure, not only the set of closed points.

7.15 Example: A Projective Scheme with No Real Points

Let

$$k = \mathbb{R}.$$

Consider

$$X = \text{Proj } \mathbb{R}[x_0, x_1]/(x_0^2 + x_1^2).$$

The real points would be

$$[a_0 : a_1] \in \mathbb{P}_{\mathbb{R}}^1$$

such that

$$a_0^2 + a_1^2 = 0.$$

Over $\mathbb{R}$, this equation has no nonzero solutions.

So

$$X(\mathbb{R}) = \emptyset.$$

But X is not necessarily the empty scheme. Over $\mathbb{C}$, the equation factors:

$$x_0^2 + x_1^2 = (x_0 + ix_1)(x_0 - ix_1).$$

So after extending scalars to $\mathbb{C}$, the scheme has two complex points.

This example shows the difference between the set of k -rational points

$$X(k)$$

and the scheme X itself.

7.16 Example: The Veronese Embedding

Let

$$S = k[x_0, x_1].$$

The degree-two homogeneous polynomials are generated by

$$x_0^2, \quad x_0x_1, \quad x_1^2.$$

Define a map

$$\mathbb{P}_k^1 \longrightarrow \mathbb{P}_k^2$$

by

$$[a_0 : a_1] \longmapsto [a_0^2 : a_0a_1 : a_1^2].$$

The image satisfies the equation

$$Y_0Y_2 - Y_1^2 = 0.$$

Thus the image is the conic

$$Y_0Y_2 = Y_1^2$$

in

$$\mathbb{P}_k^2.$$

In scheme language, this is related to the graded subring

$$k[x_0^2, x_0x_1, x_1^2] \subseteq k[x_0, x_1].$$

The Proj construction captures this projective embedding.

Thus the conic

$$\text{Proj } k[Y_0, Y_1, Y_2]/(Y_0Y_2 - Y_1^2)$$

is isomorphic to

$$\mathbb{P}_k^1.$$

7.17 Example: Projective Closure of an Affine Curve

Start with the affine parabola

$$y = x^2$$

inside

$$\mathbb{A}_k^2.$$

Its affine coordinate equation is

$$y - x^2 = 0.$$

To take the projective closure in $\mathbb{P}_k^2$, introduce homogeneous coordinates

$$[X : Y : Z].$$

On the affine chart $Z \neq 0$, we have

$$x = \frac{X}{Z}, \quad y = \frac{Y}{Z}.$$

The equation

$$y = x^2$$

becomes

$$\frac{Y}{Z} = \frac{X^2}{Z^2}.$$

Multiplying by Z^2 , we get

$$YZ = X^2.$$

Thus the projective closure is

$$\text{Proj } k[X, Y, Z]/(YZ - X^2).$$

On the affine chart $Z \neq 0$, set

$$x = \frac{X}{Z}, \quad y = \frac{Y}{Z}.$$

Then

$$YZ = X^2$$

becomes

$$y = x^2.$$

The points at infinity satisfy

$$Z = 0.$$

Plugging $Z = 0$ into

$$YZ = X^2$$

gives

$$X^2 = 0.$$

Thus

$$X = 0.$$

So the point at infinity is

$$[0 : 1 : 0].$$

Therefore the projective closure of the affine parabola adds one point at infinity.

7.18 Example: Proj of a Ring with an Unusual Grading

Let

$$S = k[x, y],$$

but give it the grading

$$\deg x = 1, \quad \deg y = 2.$$

Then

$$S_+ = (x, y).$$

This is not the standard grading of a polynomial ring in two variables.

The scheme

$$\text{Proj } S$$

is a weighted projective scheme, usually written

$$\mathbb{P}(1, 2).$$

Its points are homogeneous prime ideals not containing (x, y) , but the affine charts behave differently because x and y have different degrees.

For example,

$$D_+(x)$$

has coordinate ring

$$S_{(x)}.$$

Since

$$\deg x = 1,$$

degree-zero fractions include

$$\frac{y}{x^2}.$$

Thus

$$S_{(x)} = k \left[\frac{y}{x^2} \right].$$

On the other hand, the chart $D_+(y)$ involves degree-zero expressions such as

$$\frac{x^2}{y}.$$

This example shows that Proj is sensitive to the grading, not only to the underlying ring.

7.19 Example: Same Ring, Different Grading

Let

$$S = k[x, y].$$

With the standard grading

$$\deg x = \deg y = 1,$$

we get

$$\operatorname{Proj} S = \mathbb{P}_k^1.$$

But with the grading

$$\deg x = 1, \quad \deg y = 2,$$

we get the weighted projective scheme

$$\operatorname{Proj} S = \mathbb{P}(1, 2).$$

Therefore the symbol

$$\operatorname{Proj} S$$

depends not only on the ring S , but also on its grading.

This is very important:

Proj is a construction for graded rings, not merely for rings.

7.20 Example: The Empty Open $D_+(0)$

Let

$$S = k[x_0, x_1].$$

Take

$$f = 0.$$

Then

$$D_+(0) = \{\mathfrak{p} \in \operatorname{Proj} S : 0 \notin \mathfrak{p}\}.$$

But every prime ideal contains 0. Hence

$$D_+(0) = \emptyset.$$

Thus

$$D_+(0) = \emptyset.$$

This is the projective analogue of the affine fact

$$D(0) = \emptyset.$$

7.21 Example: The Whole Space $D_+(1)$

Since

$$1 \in S_0,$$

we have

$$D_+(1) = \{\mathfrak{p} \in \text{Proj } S : 1 \notin \mathfrak{p}\}.$$

No proper prime ideal contains 1, so

$$D_+(1) = \text{Proj } S.$$

However, 1 has degree zero, so $D_+(1)$ is usually not one of the positive-degree basic opens used to cover $\text{Proj } S$.

The important affine charts are usually

$$D_+(f),$$

where

$$f \in S_+$$

is homogeneous.

7.22 Example: Intersection of Charts in $\mathbb{P}_k^2$

Let

$$S = k[x_0, x_1, x_2].$$

Consider

$$D_+(x_0) \cap D_+(x_1).$$

By the general formula,

$$D_+(x_0) \cap D_+(x_1) = D_+(x_0x_1).$$

On the chart $D_+(x_0)$, use coordinates

$$u = \frac{x_1}{x_0}, \quad v = \frac{x_2}{x_0}.$$

The condition

$$x_1 \neq 0$$

becomes

$$u \neq 0.$$

Therefore

$$D_+(x_0) \cap D_+(x_1)$$

corresponds to

$$\text{Spec } k[u, v]_u.$$

On the chart $D_+(x_1)$, use coordinates

$$u' = \frac{x_0}{x_1}, \quad v' = \frac{x_2}{x_1}.$$

The transition functions are

$$u' = \frac{1}{u},$$

and

$$v' = \frac{x_2/x_0}{x_1/x_0} = \frac{v}{u}.$$

So the gluing rule is

$$\boxed{u' = u^{-1}, \quad v' = \frac{v}{u}.}$$

This is the usual coordinate transition between affine charts of projective space.

7.23 Recommended Example Set for Lecture Notes

For a clean lecture-note progression, the best examples are:

Example	Main point
$\text{Proj } k[x]$	Gives $\mathbb{P}_k^0$, a single point.
$\text{Proj } k[x_0, x_1]$	Gives $\mathbb{P}_k^1$, two affine charts.
$\text{Proj } k[x_0, x_1, x_2]$	Gives $\mathbb{P}_k^2$, three affine charts.
$V_+(x_2) \subseteq \mathbb{P}_k^2$	A projective line as a closed subscheme.
$\text{Proj } k[x_0, x_1, x_2]/(x_0x_2 - x_1^2)$	A projective conic.
$\text{Proj } k[x_0, x_1]/(x_1^2)$	A non-reduced projective scheme.
$\text{Proj } k$	The empty scheme.
$\text{Proj } k[x, y]$ with $\deg x = 1, \deg y = 2$	A weighted projective example.

The most important computational example is:

$$\boxed{D_+(x_i) \cong \text{Spec } k \left[\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i} \right].}$$

The most important projective closed-subscheme example is:

$$\boxed{\text{Proj } k[x_0, \dots, x_n]/I.}$$

Its closed points are

$$[a_0 : \dots : a_n]$$

satisfying

$$F(a_0, \dots, a_n) = 0$$

for every homogeneous polynomial

$$F \in I.$$

7.24 Final Summary

The examples above show the main uses of the Proj construction.

First,

$$\text{Proj } k[x_0, \dots, x_n] = \mathbb{P}_k^n.$$

Second, each standard affine chart is

$$D_+(x_i) \cong \mathbb{A}_k^n.$$

Third, homogeneous ideals define projective closed subschemes:

$$I \subseteq k[x_0, \dots, x_n] \implies \text{Proj } k[x_0, \dots, x_n]/I \subseteq \mathbb{P}_k^n.$$

Fourth, the grading matters. The same underlying ring with different gradings can give different projective schemes.

Finally, projective functions are locally represented by degree-zero homogeneous fractions:

$$\frac{F}{G}, \quad \deg F = \deg G.$$

8 Charts, Diagrams, and Tables for the Construction of Proj S

This section collects visual and tabular summaries of the construction of

$$\text{Proj } S.$$

The goal is to make the main ideas easy to remember:

$$\text{Proj } S = \{\text{homogeneous prime ideals of } S \text{ not containing } S_+\},$$

where

$$S_+ = \bigoplus_{d \geq 1} S_d$$

is the irrelevant ideal.

The central local formula is

$$\boxed{D_+(f) \cong \text{Spec } S_{(f)}}$$

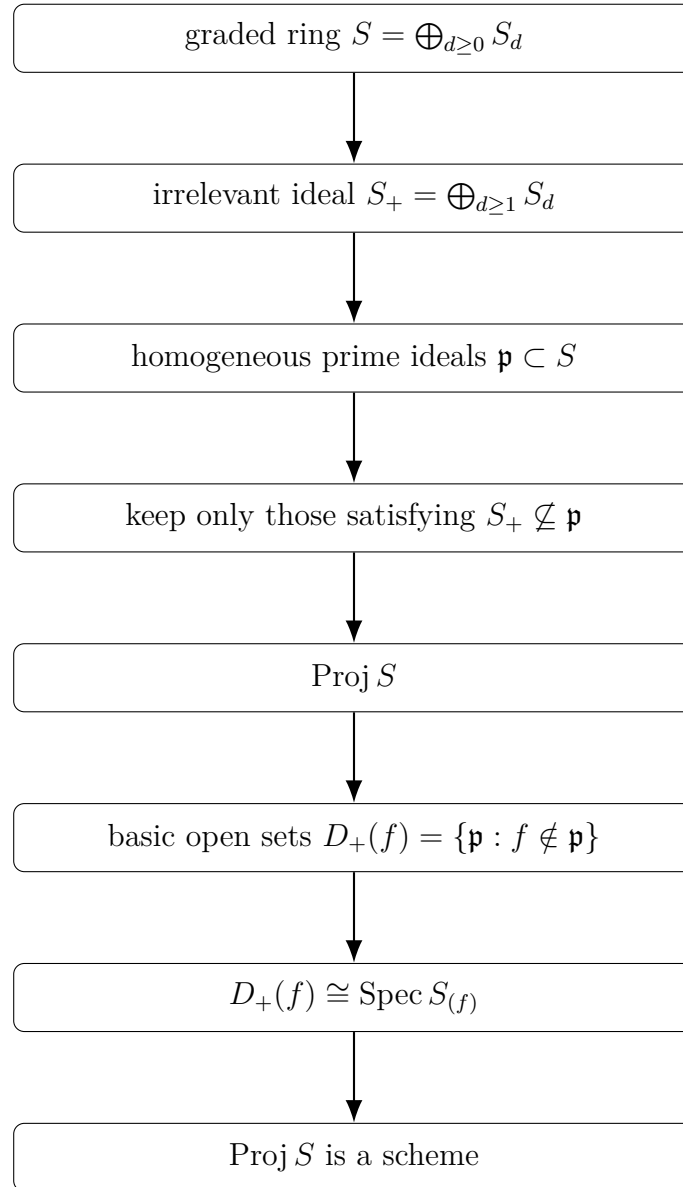
where

$$S_{(f)} = (S_f)_0.$$

Thus Proj S is studied by covering it with affine opens $D_+(f)$.

8.1 Main Construction Diagram

The construction begins with a graded ring and ends with a scheme covered by affine charts.



The most important step is

$$D_+(f) \cong \text{Spec } S_{(f)}.$$

This is what makes $\text{Proj } S$ locally affine.

8.2 Affine Versus Projective Dictionary

The construction of $\text{Proj } S$ is parallel to the construction of $\text{Spec } A$.

Affine case	Projective case
A is a ring	$S = \bigoplus_{d \geq 0} S_d$ is a graded ring
$\operatorname{Spec} A$	$\operatorname{Proj} S$
prime ideals	homogeneous prime ideals
$\mathfrak{p} \subset A$	$\mathfrak{p} \subset S$, with $S_+ \not\subset \mathfrak{p}$
$V(I)$	$V_+(I)$
$D(f)$	$D_+(f)$
A_f	$S_{(f)} = (S_f)_0$
$D(f) \cong \operatorname{Spec} A_f$	$D_+(f) \cong \operatorname{Spec} S_{(f)}$
$A_{\mathfrak{p}}$	$S_{(\mathfrak{p})}$
$\mathcal{O}_{\operatorname{Spec} A, \mathfrak{p}} \cong A_{\mathfrak{p}}$	$\mathcal{O}_{\operatorname{Proj} S, \mathfrak{p}} \cong S_{(\mathfrak{p})}$

The main difference is that in the projective case we only use homogeneous information and degree-zero fractions.

8.3 Chart Table for Projective Space

Let

$$S = k[x_0, \dots, x_n]$$

with the standard grading. Then

$$\operatorname{Proj} S = \mathbb{P}_k^n.$$

The standard affine charts are

$$D_+(x_i) = \{\mathfrak{p} \in \operatorname{Proj} S : x_i \notin \mathfrak{p}\}.$$

Chart	Condition	Coordinate ring
$D_+(x_0)$	$x_0 \neq 0$	$k \left[\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0} \right]$
$D_+(x_1)$	$x_1 \neq 0$	$k \left[\frac{x_0}{x_1}, \frac{x_2}{x_1}, \dots, \frac{x_n}{x_1} \right]$
$\vdots$	$\vdots$	$\vdots$
$D_+(x_i)$	$x_i \neq 0$	$k \left[\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i} \right]$
$\vdots$	$\vdots$	$\vdots$
$D_+(x_n)$	$x_n \neq 0$	$k \left[\frac{x_0}{x_n}, \dots, \frac{x_{n-1}}{x_n} \right]$

Therefore

$$D_+(x_i) \cong \mathbb{A}_k^n$$

and

$$\mathbb{P}_k^n = D_+(x_0) \cup D_+(x_1) \cup \dots \cup D_+(x_n).$$

8.4 Gluing Diagram for $\mathbb{P}_k^1$

For

$$\mathbb{P}_k^1 = \text{Proj } k[x_0, x_1],$$

there are two standard charts:

$$D_+(x_0) \quad \text{and} \quad D_+(x_1).$$

On $D_+(x_0)$, set

$$t = \frac{x_1}{x_0}.$$

Then

$$D_+(x_0) \cong \text{Spec } k[t].$$

On $D_+(x_1)$, set

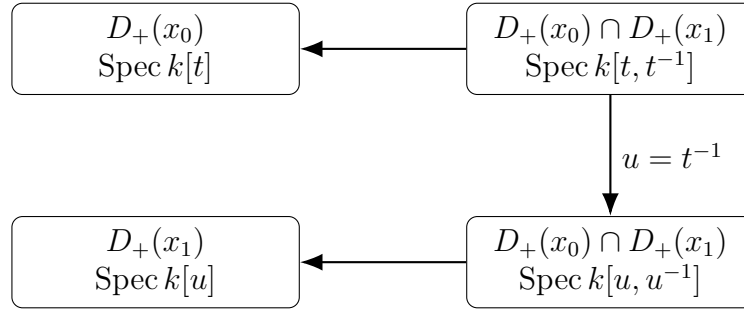
$$u = \frac{x_0}{x_1}.$$

Then

$$D_+(x_1) \cong \operatorname{Spec} k[u].$$

On the overlap,

$$u = t^{-1}.$$



Thus

$$\mathbb{P}_k^1 = \mathring{\mathbb{A}}_k^1 \cup \{\infty\}.$$

8.5 Chart-Overlap Table for $\mathbb{P}_k^2$

Let

$$\mathbb{P}_k^2 = \operatorname{Proj} k[x_0, x_1, x_2].$$

The three standard charts are:

Chart	Coordinates	Coordinate ring
$D_+(x_0)$	$u = \frac{x_1}{x_0}, \quad v = \frac{x_2}{x_0}$	$k[u, v]$
$D_+(x_1)$	$u' = \frac{x_0}{x_1}, \quad v' = \frac{x_2}{x_1}$	$k[u', v']$
$D_+(x_2)$	$r = \frac{x_0}{x_2}, \quad s = \frac{x_1}{x_2}$	$k[r, s]$

On overlaps, the transition functions are:

Overlap	Transition functions
$D_+(x_0) \cap D_+(x_1)$	$u' = \frac{1}{u}, \quad v' = \frac{v}{u}$
$D_+(x_0) \cap D_+(x_2)$	$r = \frac{1}{v}, \quad s = \frac{u}{v}$
$D_+(x_1) \cap D_+(x_2)$	$r = \frac{u'}{v'}, \quad s = \frac{1}{v'}$

This table shows explicitly how $\mathbb{P}_k^2$ is glued from three affine planes.

8.6 Diagram for Projective Closed Subschemes

Let

$$I \subseteq k[x_0, \dots, x_n]$$

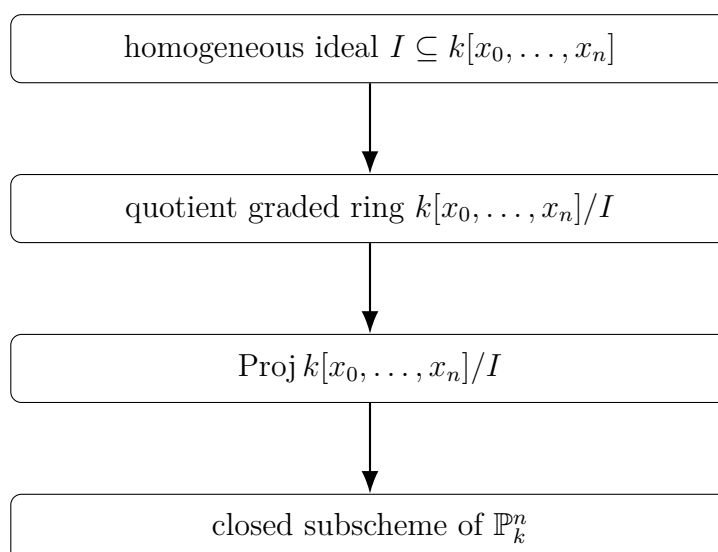
be a homogeneous ideal.

Then

$$\text{Proj } k[x_0, \dots, x_n]/I$$

is a closed subscheme of

$$\mathbb{P}_k^n.$$



The closed points are the projective classes

$$[a_0 : \cdots : a_n]$$

such that

$$F(a_0, \dots, a_n) = 0$$

for every homogeneous polynomial

$$F \in I.$$

8.7 Table of Projective Examples

Graded ring	Proj	Geometry
$k[x]$	$\text{Proj } k[x]$	$\mathbb{P}_k^0 = \text{Spec } k$
$k[x_0, x_1]$	$\text{Proj } k[x_0, x_1]$	$\mathbb{P}_k^1$
$k[x_0, x_1, x_2]$	$\text{Proj } k[x_0, x_1, x_2]$	$\mathbb{P}_k^2$
$k[x_0, \dots, x_n]$	$\text{Proj } k[x_0, \dots, x_n]$	$\mathbb{P}_k^n$
$k[x_0, x_1, x_2]/(x_2)$	$\text{Proj } k[x_0, x_1, x_2]/(x_2)$	line in $\mathbb{P}_k^2$
$k[x_0, x_1, x_2]/(x_0x_2 - x_1^2)$	$\text{Proj } k[x_0, x_1, x_2]/(x_0x_2 - x_1^2)$	projective conic
$k[x_0, x_1]/(x_1^2)$	$\text{Proj } k[x_0, x_1]/(x_1^2)$	non-reduced doubled point
k	$\text{Proj } k$	$\emptyset$

8.8 Homogeneous Versus Non-Homogeneous Equations

Projective equations must be homogeneous.

Equation	Homogeneous?	Defines projective subset?
$x_0 + x_1 = 0$	yes	yes
$x_0^2 + x_1x_2 = 0$	yes	yes
$x_0x_2 - x_1^2 = 0$	yes	yes
$x_0 + 1 = 0$	no	no
$x_0^2 + x_1 + 1 = 0$	no	no
$x_0^3 + x_1^3 + x_2^3 = 0$	yes	yes

The reason is that if F is homogeneous of degree d , then

$$F(\lambda a_0, \dots, \lambda a_n) = \lambda^d F(a_0, \dots, a_n).$$

Therefore

$$F(a_0, \dots, a_n) = 0$$

is invariant under scaling.

8.9 Degree-Zero Fraction Table

Projective regular functions are locally represented by degree-zero homogeneous fractions.

Fraction	deg numerator	deg denominator	Degree zero?
$\frac{x_1}{x_0}$	1	1	yes
$\frac{x_1^2}{x_0^2}$	2	2	yes
$\frac{x_0 x_2}{x_1^2}$	2	2	yes
$\frac{x_1^2}{x_0}$	2	1	no
$\frac{x_0 + x_1}{x_2}$	1	1	yes
$\frac{x_0^2 + x_1}{x_2^2}$	not homogeneous	2	no

The rule is:

$\frac{F}{G}$ is projectively meaningful when F, G are homogeneous and $\deg F = \deg G$.
--

8.10 Affine Charts for a Projective Conic

Let

$$X = \text{Proj } k[x_0, x_1, x_2]/(x_0 x_2 - x_1^2).$$

This is the conic

$$x_0 x_2 = x_1^2$$

in $\mathbb{P}_k^2$.

Chart	Coordinates	Affine equation
$D_+(x_0)$	$u = \frac{x_1}{x_0}, \quad v = \frac{x_2}{x_0}$	$v = u^2$
$D_+(x_1)$	$a = \frac{x_0}{x_1}, \quad b = \frac{x_2}{x_1}$	$ab = 1$
$D_+(x_2)$	$r = \frac{x_0}{x_2}, \quad s = \frac{x_1}{x_2}$	$r = s^2$

This table shows how one projective equation becomes different affine equations on different charts.

8.11 Diagram for Projective Closure of an Affine Curve

Start with the affine parabola

$$y = x^2$$

inside

$$\mathbb{A}_k^2.$$

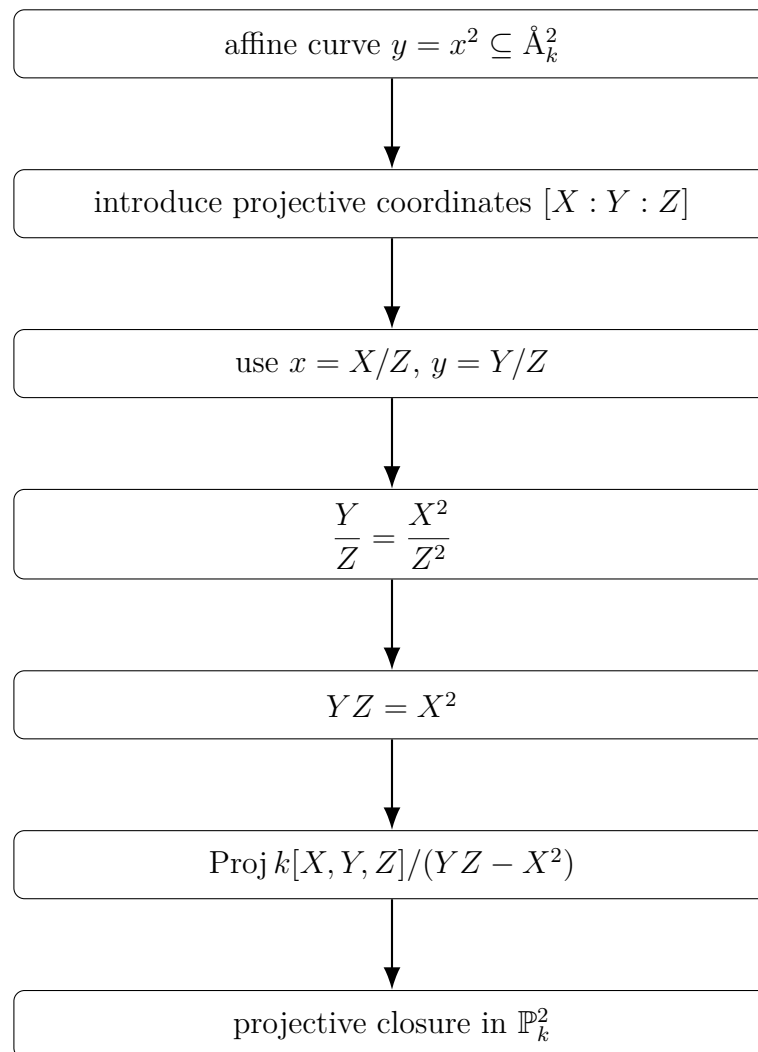
Homogenize the equation:

$$y - x^2 = 0$$

to get

$$YZ - X^2 = 0$$

in projective coordinates $[X : Y : Z]$.



The point at infinity satisfies

$$Z = 0.$$

Then

$$YZ = X^2$$

becomes

$$X^2 = 0.$$

Hence

$$X = 0.$$

Therefore the added point is

$$[0 : 1 : 0].$$

8.12 Table: Proj and Grading

The same underlying ring can give different projective geometry if the grading changes.

Ring	Grading	Geometry
$k[x, y]$	$\deg x = \deg y = 1$	$\text{Proj } k[x, y] = \mathbb{P}_k^1$
$k[x, y]$	$\deg x = 1, \deg y = 2$	$\text{Proj } k[x, y] = \mathbb{P}(1, 2)$
$k[x_0, \dots, x_n]$	$\deg x_i = 1$	$\mathbb{P}_k^n$
$k[x_0, \dots, x_n]$	$\deg x_i = w_i$	$\mathbb{P}(w_0, \dots, w_n)$

Key warning:

$\text{Proj } S$ depends on the grading of S .

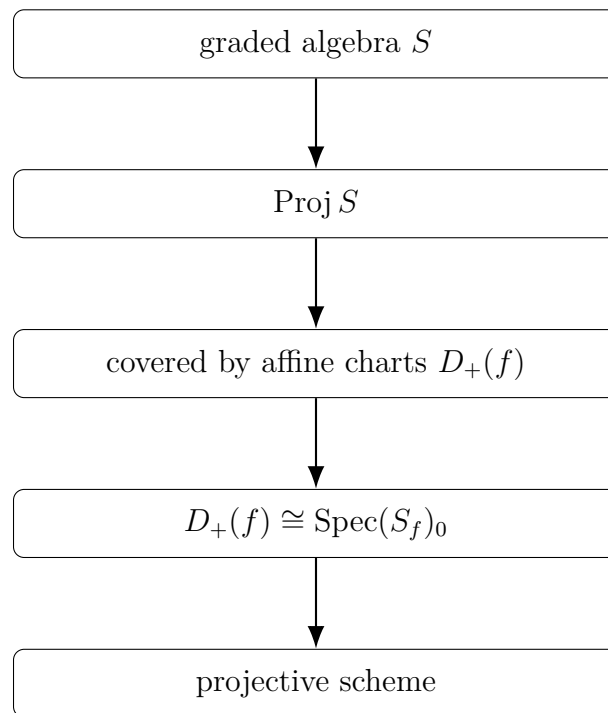
8.13 Recommended Visual Package

A clean order for lecture notes is:

Number	Visual item
1	Construction flow diagram: $S \rightarrow S_+ \rightarrow \text{Proj } S \rightarrow D_+(f) \rightarrow \text{Spec } S_{(f)}.$
2	Affine/projective dictionary table.
3	Chart table for $\mathbb{P}^n$.
4	$\mathbb{P}^1$ gluing diagram.
5	$\mathbb{P}^2$ chart-overlap transition table.
6	Projective closed subscheme diagram: $I \rightarrow S/I \rightarrow \text{Proj } S/I.$
7	Homogeneous versus non-homogeneous equation table.
8	Degree-zero fraction table.
9	Conic chart table.
10	Grading sensitivity table.

8.14 Final Summary Diagram

The whole construction can be summarized as:



The key formulas are:

$$\text{Proj } S = \{\mathfrak{p} \subset S : \mathfrak{p} \text{ homogeneous prime and } S_+ \not\subset \mathfrak{p}\}$$

and

$$D_+(f) \cong \text{Spec } S_{(f)}.$$